Climate Minsky Moments and Endogenous Financial Crises*

Matthias Kaldorf
Deutsche Bundesbank

Matthias Rottner
Deutsche Bundesbank

August 13, 2026

Abstract

How does a shift in climate policy affect financial stability? We develop a quantitative macroeconomic model with carbon taxes and endogenous financial crises to study so-called "Climate Minsky Moments". By reducing asset returns, an accelerated transition to net zero initially elevates the crisis probability substantially. However, carbon taxes enhance medium-run financial stability by diminishing the relative size of the financial sector. In the long-run, carbon taxes do not affect crisis probabilities due to directed technological change. Quantitatively, the net financial stability effect is only negative for higher social discount rates. Even then, the welfare effects of "Climate Minsky Moments" are second-order relative to the real costs and benefits of an accelerated transition.

Keywords: Climate Policy, Financial Stability, Financial Crises, Transition Risk, Non-Linearities.

JEL classification: E32, E44, G20, Q52, Q58

*matthias.kaldorf@bundesbank.de, matthias.rottner@bis.org.

We thank Markus Brunnermeier, Emanuele Campiglio, Francesca Diluiso, Sampreet Goraya, Pablo Guerron, Niko Jaakkola, Rustam Jamilov, Jochen Mankart, Givi Melkadze, Benoit Mojon, Simon Scheidegger, Hernan Seoane, Thomas Stoerk, Gauthier Vermandel, our discussants Jorge Abad, Barbara Annicchiarico, Volha Audzei, Ghassane Benmir, Daniel Monteiro, Valerio Nispi Landi, Maria Sole Pagliari, and Anna Straubinger, and participants in the ECB Conference on Financial Stability and Macroprudential Policy, CEBRA Annual Conference 2025, Annual Research Conference by the European Commission, Mannheim Quantitative Macro Workshop, NAS Macroeconomic Implications of Decarbonization Workshop, 7th annual meeting of CEBRA's IFM program, ESCB Research Clusters Climate Change and Financial Stability, Green Finance Research Advances, Czech National Bank Workshop on Financial Stability and Macroprudential Policy 2024, and the research seminars of the Bank for International Settlements, Bank of England, Deutsche Bundesbank, European Central Bank, NGFS, University of Bologna, University of St. Gallen, University of Wuerzburg and Ecole Polytechnique Paris for their comments and suggestions. The views expressed here are our own and do not necessarily reflect those of the Deutsche Bundesbank, or the Eurosystem.

1 Introduction

Does the net zero transition increase financial fragility and, if so, by how much? Answering these questions is crucial for financial regulation over the next decades, which may be characterized by a large shift away from emission-intensive technologies. Ambitious taxes on carbon dioxide emissions negatively affect the macroeconomy and asset prices by triggering a sharp and permanent drop in the productivity of emission-intensive assets. A shift in climate policy can then give rise to "Climate Minsky Moments", in which a sudden reduction in asset prices raises the concern that financial intermediaries are unable to repay depositors, triggering a financial crisis.[1] However, as no country has yet introduced sufficiently stringent climate policies, evaluating the relevance of "Climate Minsky Moments" for financial stability, the macroeconomy, and welfare using historical data is practically infeasible.

Against this background, we develop a nonlinear quantitative macroeconomic model with endogenous financial crises and carbon taxes to study the threat of "Climate Minsky Moments". Our model builds on the notion that financial intermediaries are run-prone in the spirit of Diamond and Dybvig (1983). The possibility of a systemic run on the financial system is fully endogenous and depends on the macro-financial environment. Following the seminal work by Nordhaus (2018), carbon emissions and carbon taxes affect the production sector, which faces a choice between clean and dirty technologies. Using our macro-finance-climate model, we evaluate how a climate policy that reaches net zero carbon dioxide emissions by 2050 affects financial stability and macroeconomic aggregates, both in the short- and long-run. This quantitative evaluation of endogenous financial fragility adds a novel element to the ongoing debate on financial stability in the context of climate policy and transition risk. We then examine how the threat of "Climate Minsky Moments" affects welfare and benchmark its impact against the real costs and benefits of the net zero transition.

The two key building blocks of our framework are the financial sector with endogenous financial crises and firms' technology choice, which is affected by the climate policy stance. Financial intermediaries are subject to an endogenous leverage constraint based on Adrian and Shin (2014) and Nuño and Thomas (2017). Adding risk shocks in the spirit of Christiano, Motto, and Rostagno (2014) introduces time-variation into the leverage constraint. The financial sector then faces occasional runs, driven by depositors' state-dependent willingness to roll over intermediaries' liabilities, similar to Gertler, Kiyotaki, and Prestipino (2020). Therefore, the probability of such a self-fulfilling run depends on the financial sector's leverage and the price of capital, which is directly affected by climate policy.

Climate policy enters the model through the production sector. Firms emit carbon

[1] The term "Climate Minsky Moments" was coined by Carney (2016) in the spirit of Minsky (1977).

dioxide during the production process. Emissions are taxed by the fiscal authority, but we allow for the adoption of (at least temporarily) less productive clean production technologies. Higher taxes induce firms to adopt more clean technologies, which reduces the economy's emission intensity (Nordhaus, 2008), but leads to the stranding of otherwise profitable fossil assets. Financial risks associated with asset stranding are usually referred to as *transition risk* and the implied productivity losses directly reduce welfare (Campiglio, Dietz, and Venmans, 2023). At the same time, carbon taxes increase welfare by curbing global temperature increases, which in turn depend on cumulated carbon emissions (Fernandez-Villaverde, Gillingham, and Scheidegger, 2025). In addition to these real costs and benefits of the net zero transition, climate policy has an additional welfare-relevant effect due to its impact on the frequency of financial crises.

Ambitious climate policy affects financial stability through two opposing mechanisms.[2] Increasing the carbon tax results in productivity losses directly due to the asset stranding and indirectly due to a higher carbon tax burden on firms that still use the fossil technology. At a macroeconomic level, the marginal product of capital declines, which in turn depresses the market value of assets held by the financial sector.[3] This endogenously tightens the financial sector's leverage constraint. The resulting downward pressure on asset prices elevates the likelihood of a run on the financial sector: the probability of "Climate Minsky Moments" increases. At the same time, by reducing the marginal product of capital, higher carbon taxes also reduce the incentives to accumulate capital, consistent with empirical evidence by Kaenzig (2023). Consequently, households need to absorb less capital during a downturn, which stabilizes the asset price and increases financial stability: the probability of "Climate Minsky Moments" decreases. Which of these opposing mechanisms dominates is a quantitative question.

To provide a quantitative assessment, we embed this framework in a New Keynesian general equilibrium setup. The model is calibrated to salient features of macro-financial cycles and the macroeconomic effects of climate policy. Following the directed technological change literature (Acemoglu et al., 2012; Casey, 2024), our calibration takes into account the secular shift towards emission-free technologies that is already reducing the cost of abating emissions under current carbon tax levels. To operationalize this secular

[2]Throughout most of our paper, we focus on the financial stability effects of climate policy that stem from the adoption of clean but less productive technologies. With that said, we show that the key positive and normative implications of "Climate Minsky Moments" are very similar in an extension with physical climate risk, where rising global temperatures inflict damages on the macroeconomy, following standard integrated assessment models (Nordhaus, 2008).

[3]As customary in this strand of literature, the financial sector holds capital directly and firms have no capital structure choice. Since intermediaries mostly hold loans and securities issued by non-financial firms in practice, losses on intermediaries' asset holdings should be interpreted as stemming from their loan portfolios. Empirically, Jung, Santos, and Seltzer (2024) show that ambitious climate policy induces loan portfolio losses between 1 and 6% for US banks. Berthold et al. (2023) find that carbon price increases are associated with a tightening of financial conditions and higher credit spreads in the non-financial sector.

shift, we extrapolate the historical reduction in global emission intensities observed between 1990 and 2023, which implies that net zero would be reached in 2090 even absent a further carbon tax increase. Global temperatures are linked to cumulated emissions, and we assume that they inflict utility losses on households (Stokey, 1988; Acemoglu et al., 2012; Barrage, 2019), which permits an exact welfare decomposition of the net zero transition into temperature gains, productivity losses, and "Climate Minsky Moments". Solving our model in its nonlinear specification with global methods allows us to calculate the crisis probability along different climate policy paths. Our analysis reveals that both opposing mechanisms matter, but at different time horizons.

In our main policy experiment, we compare financial stability and welfare on the current trajectory to a stringent policy path aligned with the Paris Agreement. The *Paris-aligned* transition assumes that climate policy suddenly shifts in 2025 to a steeper carbon tax path. Specifically, taxes increase linearly from their current level to the net-zero-consistent level in 2050, which is around 90 dollars per ton of carbon dioxide ($/tCO2) in our calibration. Emissions on this path imply that global surface temperatures will not increase by more than $2°C$ by the end of the century, consistent with the Paris Agreement. The sudden shift onto a steeper carbon tax path comes with a decline in intermediaries' net worth and an elevated crisis probability in the early stages of the transition. This is due to productivity losses from asset stranding and higher carbon taxes. The (annualized) crisis probability increases initially from 2% to almost 3% in the *Paris-aligned* transition.

The crisis probability then slowly declines from its early peak of 3% and falls even below the initial crisis probability of 2% to around 1.5% by 2040. Ambitious climate policy enhances financial stability in the later stages of the transition. Importantly, this does not primarily follow from an elevated financial crisis probability in the first years of the transition, which might prevent the build-up of subsequent financial fragility. Instead, the combination of high taxes and large abatement spending depresses the payoff from investment so that the capital stock is smaller. Consequently, the ratio of intermediary net worth to aggregate capital is higher, which effectively increases the financial sector's resilience to adverse risk shocks. Formally, households absorb a larger share of the capital stock without inducing large asset price drops if intermediaries face deleveraging pressures. Intuitively, our model captures a positive medium-run relationship between capital accumulation and the demand for financial services, which goes back at least to Robinson (1952), who summarized the idea as "where enterprise leads, finance follows".

In the long run, the crisis probability reverts to a level of 2%. This reflects the presence of directed technological change toward emission-free technologies. Even in the absence of carbon taxes, the economy reaches net zero at some point. Once adopting the clean technology becomes costless, the level of the carbon tax no longer affects output. As a result, the capital stock converges back to a higher long-run level, similar to the initial economy with low carbon taxes.[4]

Our non-linear model reveals that a monotonic carbon tax path can have non-monotonic financial stability effects. To determine the net financial stability effect of these opposing mechanisms, we introduce a metric of financial stability that summarizes the occurrences of crises along different transition paths. The *excess crisis probability* is defined as the discounted difference between the crisis probability under the *Paris-aligned* transition and the *current trajectory.* As in all climate policy assessments, choosing the appropriate discount rate is a highly non-trivial normative issue. We do not take a stand on the appropriate social discount rate and compute the *excess crisis probability* over a reasonable range of discount rates. The excess crisis probability is positive for sufficiently high social discount rates, as the elevated crisis probability in the short run triggered by the sudden shift to a stringent carbon tax path dominates. However, the negative financial stability impact of climate policy is limited and relatively short-lived, and the *Paris-aligned* transition path benefits earlier from a lower long-run crisis probability than the *current trajectory.* Hence, the excess crisis probability turns negative if the social discount rate is small. If future financial stability gains associated with less capital accumulation are not discounted too heavily, ambitious climate policy improves financial stability. This rejects the notion of an unambiguous trade-off between financial stability and climate policy objectives.

These results raise the quantitative question of how relevant "Climate Minsky Moments" are in comparison to the productivity losses of clean technology adoption and welfare gains from mitigating global temperature increases. Using our structural model, we benchmark the welfare relevance of "Climate Minsky Moments" against the real costs and benefits of a transition to net zero. First, we demonstrate that the welfare gains of "Climate Minsky Moments" are inversely related to the *excess crisis probability* and are positive (negative) for a low (high) social discount rate. The welfare effect of "Climate Minsky Moments" over time bears a striking resemblance to the time pattern of real costs and benefits of the transition. Second, the welfare effects of "Climate Minsky Moments" are dwarfed by the costs of switching to emission-free technologies. For example, assuming a high social discount rate of 4% p.a., "Climate Minsky Moments" reduce welfare by around 0.01% in consumption equivalents, while productivity losses from emission mitigation reduce welfare by slightly more than 1% in consumption equivalents. For a low social discount rate of 0.5% p.a., welfare increases by an additional 0.02% in consumption equivalents due to the positive long-run financial stability effects of high carbon taxes. For an intermediate social discount rate of 1.5% that is close to common estimates of long-run real discount rates, we obtain an approximate irrelevance result. "Climate Minsky Moments" do not substantially affect the key climate policy trade-off between

[4] This equivalence relies on the assumption that emission damages directly reduce utility but do not feed back into firms' incentives to accumulate capital. We relax this assumption in an extension.

productivity losses and temperature gains.

The short-run financial stability losses and medium-run financial stability gains of carbon taxes, as well as their quantitative irrelevance for welfare, are robust to several model modifications. In an extension, we model physical climate damages as a negative effect on output in the spirit of integrated assessment models (Nordhaus, 2008, Golosov et al., 2014). Thus, larger global temperature increases are associated with lower productivity and less capital accumulation. Consequently, capital and crisis probability are lower on the *current trajectory* than on the *Paris-aligned* carbon tax path. While our main results carry over to this extension, the medium- to long-run financial stability gains of ambitious climate policy are slightly diminished. This implies that "Climate Minsky Moments" are associated with welfare losses also for fairly low social discount rates. We also demonstrate robustness with respect to higher welfare losses of global warming.

Furthermore, we show that the likelihood of "Climate Minsky Moments" and their associated welfare results are robust to different climate policy modifications, such as a) clean technology subsidies, b) environmental performance standards, c) the year at which net zero is reached, and d) delayed carbon tax paths. The welfare relevance of "Climate Minsky Moments" remains at most second-order in either case. Taken together, our results push back decisively on the notion that financial stability concerns are a valid reason to delay the net zero transition, irrespective of the social discount rate. Our results instead indicate that "Climate Minsky Moments" appear to be a manageable risk, regardless of the exact specifications of the production sector, climate policy, and the climate block.

As a final step, we evaluate the potential to mitigate financial stability concerns further using the central bank's toolkit. In particular, we study the role of macroprudential policy and monetary policy through the lens of our quantitative model. Notably, the financial cycle is a crucial driver of the financial stability effects associated with stringent climate policies. Specifically, the short-run financial stability and welfare impact of stringent climate policies depend on the financial sector's loss-absorbing capacity before the switch to a Paris-aligned transition. If the financial sector is highly levered, the crisis probability jumps to a substantially higher level of around 5.5% compared to less than 3% on the average path. The reason is that the financial sector has a smaller equity buffer to absorb the losses due to asset stranding. Therefore, the build-up of macroprudential space or the implementation of climate policies during a period of strong financial fundamentals allows for the mitigation of the likelihood of "Climate Minsky Moments" substantially.

Regarding monetary policy, we show that an expansionary monetary policy stance can soften the initial impact of ambitious climate policy and the crisis probability by lowering the funding costs of the financial sector. Conversely, contractionary monetary policy exacerbates financial fragility at the onset of the transition. We model the monetary policy stance by a deterministic wedge in the interest rate rule. Contractionary monetary policy has a quantitatively greater impact for the same size of the wedge, making an overly tight monetary policy more costly than an overly loose one. This implies that monetary policy should not "lean against the transition" - as we denote a contractionary monetary policy stance in response to the shift on the Paris-aligned path. The reason is that it is too late to contain financial fragility once stringent climate policies are implemented.

Related Literature. Our paper is connected to the fast-growing literature that uses fully-fledged DSGE models with climate policy and frictions in the financial sector (e.g., Diluiso et al., 2021; Annicchiarico, Carli, and Diluiso, 2026; Carattini et al., 2024; Frankovic and Kolb, 2024; Nakov and Thomas, 2026; Airaudo, Pappa, and Seoane, 2026; Fornaro, Guerreri, and Reichlin, 2024; Dietrich, Leitenbacher, and Mueller, 2026; Sahuc, Smets, and Vermandel, 2024). Carattini, Melkadze, and Heutel (2023) use a two-sector model to show that carbon taxes can negatively affect financial intermediaries' net worth and induce a credit crunch, which motivates sector-specific macroprudential policy. Giovanardi et al. (2023) and Giovanardi and Kaldorf (2026) model idiosyncratic default and risk-taking in the firm and banking sector to study sector-specific central bank collateral policy and bank capital regulation, respectively. Barnett (2023) shows how financial frictions can give rise to fire sales of carbon-intensive assets. Campiglio, Dietz, and Venmans (2023) demonstrate that taking transition dynamics into account is pivotal in the welfare assessment of climate policies. Our paper differs from these studies that primarily focus on capital re-allocation across sectors, while abstracting from financial crises. By incorporating the possibility of runs into a climate DSGE model, we can study the systemic risk dimension of climate policy.

To incorporate the possibility of "Climate Minsky Moments", we build on the recent advancements in the macro-finance literature that incorporated endogenous financial crises in macroeconomic models, as in Brunnermeier and Sannikov (2014), Gertler and Kiyotaki (2015), Boissay, Collard, and Smets (2016), Moreira and Savov (2017), Ikeda and Matsumoto (2021), Abad, Martinez-Miera, and Suarez (2024), and Amador and Bianchi (2024). Our macro-finance model block combines pro-cyclical leverage dynamics with self-fulfilling runs, as in Gertler, Kiyotaki, and Prestipino (2020), to reconcile key features of financial cycles, following Rottner (2023). The framework features, for instance, "credit booms gone bust" dynamics (Schularick and Taylor, 2012), pro-cyclical debt issuance (Jermann and Quadrini, 2012), and the volatility paradox (Brunnermeier and Sannikov, 2014). Our paper contributes to this literature by incorporating the role of climate policy in this type of model.

We also relate to the literature on the interactions between financial stability and climate policy that uses analytically tractable models. Jondeau, Mojon, and Monnet (2021) address the risk of fire sales of emission-intensive assets. Döttling and Rola-Janicka (2025) analyze jointly optimal climate and financial policy in the context of our paper. Oehmke and Opp (2025) propose an analytically tractable two-sector model of ”green” capital requirements and find that, while green capital requirements are unable to induce the transition to net zero, they can alleviate the risk of asset stranding and thus enhance the credibility of an ambitious climate policy path. Our contribution is to build a quantitative model, which opens up the possibility of examining the impact of different climate policies on financial stability through a quantitative lens.

An alternative approach taken by several regulators is climate stress testing. Acharya et al. (2023) provide a summary and outlook for climate change stress tests. Specific examples are Arlt et al. (2025), Jung, Santos, and Seltzer (2024) and IMF (2022), who evaluate the effects of specific climate change scenarios on banks' loan portfolios for the euro area, US and UK, respectively. Chabot and Bertrand (2023) empirically investigate the effects of physical climate and transition risk on individual European banks and at the systemic level. Brunetti et al. (2022) review climate stress test results conducted in several jurisdictions. By design, stress tests cannot uncover positive financial stability effects of the transition, as they abstract from the financial sector's deleveraging incentives along the net zero transition. In contrast, by taking a general equilibrium perspective, we can fully endogeneize the responses of the different economic agents during the transition to net zero. It also opens up the possibility of a welfare analysis to benchmark the welfare impact of “Climate Minsky Moments” against costs and benefits of the net zero transition in the real sector.

2 Model

The model consists of three building blocks related to carbon emissions and climate policy, the financial sector, and the macroeconomy, respectively. Following the macro-climate literature (Acemoglu et al., 2012; Golosov et al., 2014) , firms can either operate a dirty technology, which is associated with carbon dioxide emissions that contribute to global temperature increases and are subject to carbon taxes, or a less productive clean technology that allows firms to reduce their carbon tax bill. The financial sector embeds an endogenous leverage constraint (Adrian and Shin, 2014; Nuño and Thomas, 2017) in a model with endogenous financial crises as in Gertler, Kiyotaki, and Prestipino (2020), following the setup of Rottner (2023). This representation of financial fragility and climate policy is embedded into an otherwise standard New Keynesian setup, which turns out to improve the model's fit to macro-financial dynamics.

Production Technology and Climate Policy We first describe the production sector, which is the part of the model where carbon dioxide emissions and taxes enter. Firms are competitive and should be interpreted as multi-product conglomerates in the spirit of Blanco et al. (2024). Each firm produces a mass-one continuum of goods that is indexed by $\eta \in (0, 1)$. Goods varieties are produced with an identical Cobb-Douglas production function $Y_t = AK_{t-1}^{\alpha} L_t^{1-\alpha}$ using capital K_t and labor L_t as inputs. For each variety η, firms can choose to use a dirty technology that emits one unit of carbon dioxide for each unit of output during the production process or a clean technology that is emission-free.

Carbon dioxide emissions are taxed at the (potentially time-varying) rate τ_t^c, such that the after-tax revenues for a dirty goods variety are given by $(1 - \tau_t^c)Y_{\eta,t}$. In order to reduce emissions, firms can adopt a clean technology for each good that is - at least temporarily - less productive. To capture the idea that some goods can be produced with clean technologies more easily, we assume that the productivity loss can be written as a convex function $b_{1,t}\eta^{b_2}$ over the space of goods, where $b_{1,t} \geq 0$ and $b_2 > 1$. This convex form of productivity losses implies that it is almost costless to switch for some goods, such as services with a very small emission intensity, while it becomes very costly to operate in an emission-free fashion for other goods varieties, e.g., manufacturing involving cement or steel.[5] Furthermore, the time dependence of the slope parameter captures technological advances related to emission-free production processes.

In this setting, carbon taxes determine the share of goods that are produced with a clean technology. As the after-tax revenues of selling such goods are given by $(1 - b_1\eta^{b_2})Y_{\eta,t}$, all goods varieties satisfying $b_{1,t}\eta^{b_2} < \tau_t^c$ will be produced in an emission-free fashion. We can pin down the marginal variety η_t^* for which firms are indifferent:

$$\eta_t^* = \min\left\{ \left(\frac{\tau_t^c}{b_{1,t}}\right)^{\frac{1}{b_2}}, 1 \right\} . \tag{1}$$

Consequently, η_t^* is the (time-varying) share of goods produced in the emission-free technology. The clean product share η_t^* increases with the carbon tax. The min-operator captures the idea that all goods are produced with the clean technology if the tax exceeds $\tau_t^c > b_{1,t}$, and we refer to such taxes as consistent with "net zero" carbon dioxide emissions.

The aggregate productivity loss associated with asset stranding obtains from integrating the productivity loss for all goods that use clean technologies, $\int_0^{\eta_t^*} b_{1,t}\eta^{b_2} Y_t d\eta = \frac{b_{1,t}}{b_2+1}(\eta_t^*)^{b_2+1} Y_t$. This term enters the resource constraint. Aggregate emissions are therefore given by $(1 - \eta_t^*)Y_t$, while the aggregate carbon tax paid by all firms in period t is

[5] One advantage of our formulation with a choice between clean and dirty technologies is that it admits a net zero emission steady state, which is essential for our long-term analysis. A common modelling assumption in the context of asset stranding is a CES formulation with a large elasticity of substitution, but without allowing for technological choice. Without further modifications, this approach precludes a net zero emissions scenario, as some degree of dirty capital is irreplaceable under reasonable calibrations. Our macroeconomic model with heterogeneous production technologies instead relies on the idea that some goods are very emission-intensive but are also very difficult to replace with clean production processes. Firms will operate the dirty technology even under high carbon taxes and pay the tax. This still reduces the return on assets even though these assets do not "strand" in a narrow sense.

given by $\int_{\eta_t^*}^{1} \tau_t^c Y_t d\eta = \tau_t^c(1-\eta_t^*)Y_t$. Carbon tax revenues are rebated to households in a lump sum fashion. We consider a modification in which carbon tax revenues subsidize firms' adoption of clean technologies.

To link climate policy directly to the accumulation and pricing of capital in our model economy, it is helpful to define the policy-induced return on capital wedge B_t, which summarizes the productivity losses from carbon taxation and asset stranding per unit of aggregate output:

$$B_t \equiv \tau_t^c\,(1-\eta_t^*) + \frac{b_{1,t}}{b_2+1}\,(\eta_t^*)^{b_2+1}\ . \tag{2}$$

We allow for technological change directed at emission-free technologies, which we model as a secular decline in the productivity loss parameter $b_{1,t}$ over time, such that it eventually is costless to operate emission-free technologies. However, it is important that at least the adoption of clean technologies entails a productivity loss in the medium run, which appears to be a mild assumption.

The realized return on investment is given by $R_t^K = [(1-\delta)Q_t + Z_t]/Q_{t-1}$. Taking the price of the intermediate good p_t as given, the first-order conditions for capital and labor can be expressed as:

$$Z_t = (p_t - B_t)\alpha\frac{Y_t}{K_{t-1}}, \quad \text{and} \quad W_t = (p_t - B_t)(1-\alpha)\frac{Y_t}{L_t}\ . \tag{3}$$

Since emissions are proportional to total output, climate policy does not affect the aggregate capital share but depresses total factor productivity.

Emissions, Temperature and Climate Damages Emissions are linked to economic damages through increases in global temperatures. Following the climate economics literature, carbon emissions accumulate into a stock of atmospheric carbon dioxide E_t, which we can map into temperature changes (Fernandez-Villaverde, Gillingham, and Scheidegger, 2025). Specifically, the global temperature change relative to pre-industrial levels, $\Delta\mathcal{T}_t \equiv \mathcal{T}_t - \mathcal{T}_{1850}$, is positively related to cumulated emissions $E_t \equiv E_{2024Q4} + \sum_{s=2025Q1}^{t} e_s$, where E_{2024Q4} corresponds to cumulated emissions up to the last period before the (potential) policy change. While DSGE models are usually expressed in per capita terms, global emissions need to take population growth into account. We do so by accounting for simple population growth rates, using $e_t = (1-\eta_t)Y_t(1+g*(t-T_0))$ for $t > T_0 = 2024Q4$. Following Fernandez-Villaverde, Gillingham, and Scheidegger (2025), we approximate this relationship by:

$$\Delta\mathcal{T}_t = \Psi_{CCR}E_t\ . \tag{4}$$

To map global temperature increases into economic damages, we assume that temperature changes inflict a utility loss $\mathcal{D}(\Delta\mathcal{T}_t)$ as in Stokey (1988), Acemoglu et al. (2012); or Barrage (2019). Utility losses are quadratic in the global temperature increase relative to pre-industrial levels:

$$\mathcal{D}(\Delta\mathcal{T}_t) = \gamma_D(\Delta\mathcal{T}_t)^2. \tag{5}$$

Such quadratic loss functions are commonly used in integrated assessment models, for example Nordhaus (2008). The key advantage of this specification is that it allows us to cleanly compare the welfare costs of "Climate Minsky Moments" relative to other benefits and costs of curbing global warming without introducing additional interactions between the climate block and the production sector. Temperature changes do not affect any equilibrium outcome, but they affect welfare. We relax this assumption in an extension with physical risk that affects productivity in the real sector. The extension introduces feedback between the global climate and macroeconomic outcomes. Temperature increases are mapped to a damage function that lowers output instead of inflicting utility losses.

Households The representative household consists of workers and managers that have perfect insurance for their consumption C_t. Workers supply labor L_t and earn the wage W_t. Managers run financial intermediaries that return their net worth to the household with a probability of $1-\theta$. New intermediaries enter each period and receive a transfer from the household, who owns non-financial firms and receives their profits. The variable T_t captures all transfers from the public sector.

Households save in terms of one-period deposits D_t, which promise to pay the gross interest rate of $\overline{R}_t^D$ next period. However, in case of a run, households receive only the fraction $x_t^\star$ of the promised return, which we refer to as the recovery ratio. The realized gross return R_t^D depends on the realization of a run in period t:

$$R_t^D = \begin{cases} \overline{R}_{t-1}^D & \text{if no run takes place in period } t \,, \\ x_t^\star \overline{R}_{t-1}^D & \text{if a run takes place in period } t \,, \end{cases} \tag{6}$$

where x_t^* is the recovery rate on deposits, which we derive below. Additionally, households and intermediaries can invest in the production sector by purchasing capital K_t^H and K_t^B, respectively, that give them ownership in the intermediate good firm. The rental rate on capital is denoted by Z_t, while its market price is denoted by Q_t. Total end-of-period securities are given by $K_t = K_t^H + K_t^B$. Households maximize utility subject to the following period budget constraint:

$$C_t = W_t L_t + D_{t-1}R_t^D - D_t + T_t - Q_t K_t^H + \big(Z_t + (1-\delta)Q_t\big)K_{t-1}^H \,. \tag{7}$$

We assume that households are less efficient in managing securities than intermediaries (Brunnermeier and Sannikov, 2014). As in Gertler, Kiyotaki, and Prestipino (2020), households incur a utility cost from managing capital $\mathcal{C}(K_t^H, K_t)$. As discussed before, damages associated with global temperature increases since pre-industrial times $\mathcal{D}(\Delta\mathcal{T}_t)$ enter the utility function directly. The otherwise standard period utility function is given by:

$$u(C_t, L_t, \frac{K_t^H}{K_t}, K_t, \Delta\mathcal{T}_t) = \frac{C_t^{1-\gamma_C}}{1-\gamma_C} - \frac{L_t^{1+\gamma_L}}{1+\gamma_L} - \mathcal{C}(\frac{K_t^H}{K_t}, K_t) - \mathcal{D}(\Delta\mathcal{T}_t) \ . \tag{8}$$

Capital Management Costs We specify households' capital management costs as

$$\mathcal{C}\Big(\frac{K_t^H}{K_t}, K_t, A_t\Big) = \frac{\omega_F}{2}\Big(\frac{K_t^H}{K_t} - \gamma^F\Big)^2 K_t \frac{A}{A_t} \ . \tag{9}$$

Here $A_t \equiv A(1 - \frac{b_{1,t}}{b_2+1}(\eta_t^*)^{b_2+1})$ denotes the *effective* aggregate productivity, which consists of exogenous total factor productivity A and the abatement costs spent per unit of output. De-trending the capital management cost function ensures that the long-run crisis probability is independent of secular productivity trends. Some properties of this cost function deserve attention, as they turn out to be important in the quantitative analysis. We focus on the case where households hold more capital than they prefer to ($\frac{K_t^H}{K_t} > k^H$), which holds in all states throughout our numerical simulation. Holding aggregate capital K_t and effective productivity A_t constant, management costs increase in the household capital share ($\frac{\partial\mathcal{C}}{\partial\frac{K_t^H}{K_t}} > 0$) and up to the level $\frac{\omega_F}{2}(1-\gamma^F)^2 K_t$ at which households manage the entire capital stock. Holding the household capital share $\frac{K_t^H}{K_t}$ constant, management costs also increase in aggregate capital. With both partial derivatives being positive, its cross-derivative

$$\frac{\partial^2\mathcal{C}}{\partial\frac{K_t^H}{K_t}\partial K_t} = \omega^F\Big(\frac{K_t^H}{K_t} - \gamma^F\Big) \ , \tag{10}$$

is also positive in the numerically relevant region $\frac{K_t^H}{K_t} > \gamma^F$.

These features of the management cost function are consistent with the idea that monitoring investment projects is more costly to households than financial intermediaries, and this cost disadvantage is larger if the capital stock is large due to economies of scale in monitoring technologies (Diamond, 1984). Put differently, financial intermediation services are in higher demand if the capital stock is large. This notion is in line with a large literature studying the finance-growth nexus. For instance, a larger number of investment projects increases the benefit from delegated monitoring of investment projects (Blackburn and Hung, 1998) or incentivizes market entry of financial intermediaries with positive cost effects due to specialization and competitiveness (Sussman, 1993). More generally, the positive relationship between capital accumulation and the demand for financial services goes back at least to Robinson (1952) who summarized the idea as "where enterprise leads, finance follows". Reviews on the theoretical and empirical literature are provided by Pagano (1993) and Levine (1997), respectively.[6]

While the cost function (9) has intuitively appealing features that are in line with the finance-growth literature, it bears implications for the financial stability effects of the net zero transition. The positive cross-derivative implies that capital management costs increase more strongly in response to deleveraging in the financial sector if the economy has accumulated a large amount of capital. Put differently, the capital demand function is steeper. The associated capital price drop is then more pronounced, thereby increasing the likelihood of systemic financial crises. Lastly, it is important to keep in mind that the (unobservable) empirical counterpart to capital management costs has to be corrected for secular trends, such as long-term productivity or population growth. If this were not the case, the model would (counter-factually) predict a crisis probability close to zero fifty years ago, when the capital stock was much smaller. Instead, one should think about long-term growth affecting the capital management costs in the same way as all macroeconomic aggregates. By contrast, ambitious climate policy and the implied sectoral re-allocation are associated with at least a temporary drop of aggregate productivity below trend. We revisit the role of technological productivity in a model extension.

Financial Intermediaries and Risk-Shifting Incentives Financial intermediaries convert their capital holdings into ω_t efficiency units, either using a safe or a risky technology. While the safe technology converts capital into one efficiency unit ($\omega_t = 1$), the risky technology is subject to idiosyncratic productivity shocks $\widetilde{\omega}$. The shock is i.i.d. over time and intermediaries and follows a log-normal distribution:

$$\log \widetilde{\omega}_t \overset{iid}{\sim} N\left(\frac{-\xi_t^2 - \psi}{2}, \xi_t\right) , \tag{11}$$

where $\psi < 1$. The good technology is superior as it has a higher mean and a lower variance due to $\psi < 1$ (see also Adrian and Shin, 2014 and Nuño and Thomas, 2017). The risky technology is characterized by higher upside risk due to the possibility of a large idiosyncratic shock realization $\widetilde{\omega}$. The volatility of idiosyncratic productivity shock ξ_t is an exogenous driver of financial cycles and an important trigger of financial crises in this model. In the spirit of Christiano, Motto, and Rostagno (2014), ξ_t is exogenous

[6] It is worth noting that writing the cost function in terms of household and total capital holdings $\mathcal{C}'(K_t^H, K_t)$ implies a negative partial derivative ($\frac{\partial \mathcal{C}'}{\partial K_t} < 0$) that reflects the idea that managing K_t^H units of capital is less costly to households in deep financial markets with plenty of investment opportunities, which allow for more portfolio diversification (Saint-Paul, 1992 or Acemoglu and Zilibotti, 1997).

and follows an AR(1) process:

$$\xi_t = (1-\rho^\xi)\xi + \rho^\xi \xi_{t-1} + \sigma^\xi \epsilon_t^\xi, \quad \text{where } \epsilon_t^\xi \sim N(0,1)\ . \tag{12}$$

The intermediary earns the return $R_t^{K,j}$ which depends on the stochastic aggregate return R_t^K and (potentially) also on the realized idiosyncratic shock realization if the intermediary invested into the risky technology $\widetilde{\omega}_t^j R_t^K$. The aggregate return depends on the price of capital Q_t and the profits per unit of capital $R_t^K = [(1-\delta)Q_t + Z_t]/Q_{t-1}$. The threshold realization $\overline{\omega}_t^j$ where the intermediary can exactly cover the face value of the deposits is given by

$$\overline{\omega}_t^j = \frac{\overline{R}_{t-1}^D D_{t-1}^j}{R_t^K Q_{t-1} K_{t-1}^{Bj}}\ . \tag{13}$$

Limited liability protects the intermediary in case of default, which distorts the intermediary's technology choice. The intermediary declares bankruptcy if the productivity shock realization is below $\overline{\omega}_t^j$. In this case, households seize the intermediaries' assets instead of receiving the promised deposit value. For this reason, the intermediary has an incentive to invest in the risky technology. Intermediaries profit fully from the upside risk, while limited liability eliminates the downside risk. The gain from limited liability for the risky technology is:

$$\Omega_t^j = \int_{-\infty}^{\overline{\omega}_{t+1}^j} (\overline{\omega}_{t+1}^j - \widetilde{\omega}) dF_t(\widetilde{\omega}) > 0. \tag{14}$$

In contrast, the gain from limited liability due to idiosyncratic risk is zero for the good technology. This creates a trade-off between the good technologies' higher mean return and the gains from limited liability for risky technology.

This results in a maximization problem, in which the financial intermediary maximizes the franchise value subject to an incentive and participation constraint. While we summarize the problem here, the complete derivation is relegated to Appendix A.1. The incentive constraint ensuring that intermediaries only invest in the good technology enters as an additional equilibrium condition:

$$(1-\pi_t)\mathbb{E}_t^N\left[\Lambda_{t,t+1}R_{t+1}^K(\theta\lambda_{t+1}^j+(1-\theta))\left(1-e^{\frac{-\psi}{2}}-\Omega_{t+1}^j\right)\right] = \pi_t\mathbb{E}_t^R\left[\Lambda_{t,t+1}R_{t+1}^K\left(e^{-\frac{\psi}{2}}-\overline{\omega}_{t+1}^j+\Omega_{t+1}^j\right)\right]\ . \tag{15}$$

where π_t is the probability of a run in period $t+1$. The expectation operators $\mathbb{E}_t^N[\cdot]$ and $\mathbb{E}_t^R[\cdot]$ are conditioned on the absence of a run and the occurrence of a run in period $t+1$, respectively. The trade-off between a higher mean return $(1-e^{\frac{-\psi}{2}})$ and the upside risk Ω_{t+1}^j can be seen on the LHS. In the case of a run, there is an additional gain of investing in the risky technology, as displayed on the RHS. The risky technology offers the possibility of having positive net worth despite a run if the idiosyncratic shock exceeds $\widetilde{\omega}_t^i > \overline{\omega}_t$.

The variable λ_t^j on the LHS of eq. (15) is the multiplier on intermediaries' participation constraint, which we derive next. The return on deposits needs to be sufficient such that households are willing to provide deposits to intermediaries. While households earn the predetermined interest rate $\overline{R}_t^D$ in normal times, households recover the gross return on capital if a run occurs. As the return on deposits in a run is lower, an increase in the run probability π_t increases intermediaries' funding costs. The participation constraint can be written as:

$$(1-\pi_t)\mathbb{E}_t^N\left[\beta\Lambda_{t,t+1}\overline{R}_t^D D_t^j\right] + \pi_t\mathbb{E}_t^R\left[\beta\Lambda_{t,t+1}R_{t+1}^K Q_t K_t^{Bj}\right] = D_t^j \,. \tag{16}$$

Note that it turns out that the incentive constraint and the participation constraint do not depend on intermediary-j-specific values, so we can drop the subscript j. Thus, we can sum up over the intermediaries to obtain aggregate values.

Runs and Equilibrium Selection In our model, a systemic financial crisis corresponds to a state where households are unwilling to roll over deposits. Runs are self-fulfilling in the sense that households' expectations about a low liquidation value of financial intermediaries' assets induce them to withdraw deposits, which forces intermediaries to sell assets at fire sale prices, justifying households' expectations. The systemic nature of runs in our model is reflected by the idea that it destroys the entire net worth of the financial system, i.e., $N_{S,t} = 0$. Newly entering intermediaries and households are the only agents left to acquire assets, which induces the price of capital to fall dramatically. We denote the fire sale price of capital by Q_t^* to determine whether a self-fulfilling run is supported and define the recovery ratio

$$x_{t+1}^* \equiv \frac{\big((1-\delta)Q_t^* + Z_t\big)K_{t-1}^B}{\overline{R}_{t-1}D_{t-1}} \,. \tag{17}$$

Runs are possible if $x_{t+1}^* < 1$ (Gertler and Kiyotaki, 2015). If the run equilibrium is possible, we select equilibria using a sunspot shock ϵ_t^π, following Cole and Kehoe (2000). The sunspot shock takes the value one with probability Υ and zero otherwise. The run probability follows as

$$\pi_t = \text{prob}(x_{t+1}^* < 1)\cdot\Upsilon \,. \tag{18}$$

Note that our model abstracts from deposit insurance, which is arguably less relevant in the context of systemic financial crises than for idiosyncratic bank failures, since the deposit insurance's credibility and ability to pay might itself be questionable in a systemic financial crisis. However, the outlined mechanism behind a financial crisis and multiple equilibria is robust as long as deposit insurance is imperfect, e.g., deposits are only partially protected or bank failure is associated with deadweight costs, which is a common assumption in the literature, see Begenau and Landvoigt (2022) and Elenev, Landvoigt, and Van Nieuwerburgh (2021) among others. The reason is that withdrawal is still preferable from an individual household perspective in the presence of imperfect deposit insurance.

Closing the model In the remaining part, we close the model by describing the retailers, the investment goods producers, and monetary policy. Monopolistically competitive retail good firms buy the intermediate goods and transform them into a differentiated final good Y_t^j. Households' final good bundle Y_t, which is given by a CES-aggregate over all final goods varieties, and the price index are:

$$Y_t = \left[\int_0^1 (Y_t^j)^{\frac{\epsilon-1}{\epsilon}} dj\right]^{\frac{\epsilon}{\epsilon-1}}, \quad \text{and} \quad P_t = \left[\int_0^1 (P_t^j)^{1-\epsilon} dj\right]^{\frac{1}{1-\epsilon}}, \tag{19}$$

where the demand for the final good variety j negatively depends on its relative price:

$$Y_t^j = \left(P_t^j/P_t\right)^{-\epsilon} Y_t \; . \tag{20}$$

Retailers set prices to maximize profits subject to Rotemberg price adjustment costs:

$$\mathbb{E}_t \sum_{s=0}^{\infty} \Lambda_{t,t+s} \left[\Big(\frac{P_{t+s}^j}{P_{t+s}} - \overline{p}_{t+s}\Big) Y_{t+s}^j - \frac{\rho^r}{2} Y_{t+s}^j \Big(\frac{P_{t+s}^j}{\Pi P_{t+s-1}^j} - 1\Big)^2\right], \tag{21}$$

where Π is the inflation target set by the central bank. Since their production function is linear in the intermediate good, retailers' marginal cost are simply given by the price of the intermediate good $MC_t = \overline{p}_t$. The New Keynesian Phillips curve follows as:

$$\left(\frac{\Pi_t}{\Pi} - 1\right)\frac{\Pi_t}{\Pi} = \frac{\epsilon}{\rho^r}\Big(MC_t - \frac{\epsilon-1}{\epsilon}\Big) + \Lambda_{t,t+1}\left(\frac{\Pi_{t+1}}{\Pi} - 1\right)\frac{\Pi_{t+1}}{\Pi}\frac{Y_{t+1}}{Y_t} \; . \tag{22}$$

Investment good producers transform I_t units of the final good into $(a_1(I_t/K_{t-1})^{1-a_2} + a_0)K_{t-1}$ units of the investment good, which they sell at price Q_t. Solving the maximization problem

$$\max_{I_t} \; Q_t \left(a_1(I_t/K_{t-1})^{1-a_2} + a_0\right) K_{t-1} - I_t \; , \tag{23}$$

yields an investment good supply function. The law of motion for capital is given by $K_t = (1-\delta)K_{t-1} + \Gamma\left(I_t/K_{t-1}\right)K_{t-1}$.

The monetary authority sets the interest rate R_t^I using a Taylor Rule subject to the zero lower bound:

$$R_t^I = \max\left\{R^I\left(\frac{\Pi_t}{\Pi}\right)^{\kappa_\Pi}\left(\frac{MC_t}{MC}\right)^{\kappa_y}, 1\right\}\ , \tag{24}$$

where deviations of marginal costs from their deterministic steady state MC reflect the output gap. To connect this rate to the household, there exists a one-period bond in zero net supply that pays the riskless nominal rate R_t^I. The associated Euler equation governs the pass-through from the monetary policy rate to the macroeconomy:

$$\mathbb{E}_t\left[\Lambda_{t,t+1}R_t^I/\Pi_{t+1}\right] = 1\ . \tag{25}$$

The resource constraint includes the price adjustment and productivity losses from using the clean technology:

$$Y_t = C_t + I_t + G + \frac{\rho^r}{2}\left(\frac{\Pi_t}{\Pi} - 1\right)^2 Y_t + \frac{b_{1,t}}{b_2+1}\left(\frac{\tau_t^c}{b_{1,t}}\right)^{\frac{b_2+1}{b_2}} Y_t\ , \tag{26}$$

where G is government spending.

3 Calibration and Solution

3.1 Parameter Choices

Each period corresponds to one quarter. We parameterize our model to match salient features of the macroeconomy, the financial sector, and climate policy. This results in a general calibration strategy that can easily be adapted to specific countries. When we target specific moments unrelated to the climate block, we compute the model-implied moments in the economy without carbon taxes. An overview of the parameterization is given in Table 1.

Conventional parameters The discount factor β is chosen to account for an average risk-free rate of 1.0% in an economy without carbon taxes.[7] The labor supply curvature $\gamma_L = 4/3$ implies a Frisch labor elasticity of 0.75 following Chetty et al. (2011), while we use log utility for consumption ($\gamma^C = 1$). We normalize output via the long-run TFP level A and target a government spending-to-output ratio of 20%. The capital share α is set to 0.33 and the depreciation rate δ to 0.025. The Rotemberg pricing parameter ρ^r is set to 178, implying a duration of 5 quarters in the related Calvo framework. The investment adjustment cost parameters a_0 and a_1 are set to normalize the asset price and

[7]Carbon taxes have a negligible effect on the risk-free rate in this model.

investment output. The curvature of the investment adjustment cost parameter is set in line with Bernanke, Gertler, and Gilchrist (1999). The central bank targets an inflation rate of 2%, while the responses to the output gap $\kappa_y = 0.125$ and inflation $\kappa_\pi = 2.0$ are set to conventional choices.

Climate block The parameters related to the climate policy block of the model are set to match key properties of carbon emissions and the macroeconomic impact of carbon taxes. We exploit that the cost of adapting and operating the clean technology map directly into the abatement cost function used in the DICE model of Nordhaus (2018). Its curvature is set to $b_2 = 1.6$, which is a conventional value in the literature. Technological change directed toward emission-reduction technologies could reduce the costs of operating clean technologies permanently, see Acemoglu et al. (2012), Fried (2018), Hassler, Krusell, and Olovsson (2021), Casey (2024), among others. We incorporate this concept in our framework by assuming that the cost parameter $b_{1,t}$ declines linearly over time from the starting value $b_{1,1990} = 0.075$ in 1990 to $b_{1,2090} = 0$ in the long run. The values are obtained by extrapolating the linear decline of global emission intensities by 33% from 1990 to 2024, which corresponds to 1% per year. Hence, we impose that abatement costs reach zero in $T_{max} = 2090$. Linearly interpolating the cost parameters, $b_{1,t} = t\frac{b_{1,1990} - b1,2090}{T_{max} - T_{ini}}$, we get a $b_{1,2025} = 0.05$. This value implies that 1.9% of GDP would be spent on emission abatement if net zero taxes were implemented immediately, consistent with commonly used integrated assessment models (Nordhaus, 2018) and related E-DSGE models (Heutel, 2012).

As a next step, we need to parameterize the impact of emissions on temperature and economic losses. The population growth rate is set to 0.0025 per quarter, in line with the current global population growth. The impact of cumulative carbon emissions on temperature changes relative to pre-industrial levels is captured by the parameter Ψ_{CCR}. Empirically, this parameter lies in the range of $0.27° - 0.63°$ per 1,000 GtCO2 emitted (Fernandez-Villaverde, Gillingham, and Scheidegger, 2025). We pick $\Psi_{CCR} = 0.58°$ to reconcile the global temperature increases since pre-industrial levels: cumulated global emissions currently amount to $E_{2024Q4} = 2{,}400$ GtCO2, such that we can back out the empirically observed temperature increase $\Delta\mathcal{T}_t = \Psi_{CCR} \cdot \frac{2400}{1000} = 1.4°\text{C}$ for 2024. To put these numbers into perspective, note that under a constant carbon tax of 5$/tCO2, cumulated emissions in 2100 would amount to around 4,700 GtCO2, implying a global temperature increase of slightly more than 2.7°.

The effect of global temperature increases on economic damages is subject to considerable uncertainty, ranging from small GDP losses of around 3% in earlier integrated assessment models (Nordhaus, 2008) to around 30% (Bilal and Kaenzig, 2026). In our calibration, we draw on recent empirical work by Nath, Ramey, and Klenow (2024) and express GDP losses as $\Delta Y_t = -\gamma_T \Delta\mathcal{T}_t$. Using temperature shocks at the country level, Nath, Ramey, and Klenow (2024) estimate that a global temperature increase by $\Delta\mathcal{T}_t = 3.7°$ by 2100 is associated with a GDP loss of 7-12%. Using the mid-point of 9.5%, this implies a GDP loss of $\gamma_T = 2.5\%$ for each degree of global warming. This value is slightly larger, but of similar magnitude to findings of Caggese et al. (2024), who combine firm-level data with granular temperature shocks and estimate that a $4°C$ warming scenario induces a GDP loss of around 7%. We obtain $\gamma_D = 0.0095$ for the quadratic utility loss function associated with global temperature increases in equation (8).[8]

Financial sector parameters The financial sector parameters are set to target salient features of financial cycles and systemic financial crises. We target an intermediary asset share of 1/3, implying that one-third of securities are funded by runnable deposits. For this reason, we set the target share of the household's asset holdings to $\gamma^F = 0.38$. The leverage of the financial intermediaries is set to 15, in line with equity to capital holdings in the financial sector of 6.67%. This value is obtained by setting households' intermediation cost to $\omega_F = 0.045$. The parameter controlling the mean return of the risky of technology follows Rottner (2023). The intermediary survival probability is set to a rather low value of $\zeta = 0.885$, which is helpful to incorporate runs in this type of model and is in line with the credit spread of 90 basis points over the risk-free rate (Gertler and Kiyotaki, 2015). The parameter that governs the initial endowment to new intermediaries is implied by the other parameters of the model. We set the standard deviation of our risk shock to match an annual run frequency of 2%, a value that is well in line with the evidence on financial crises in the macrohistory database of Jordà, Schularick, and Taylor (2017). The persistence of the shock follows Rottner (2023). The probability governing the sunspot shock is normalized to $\Upsilon = 0.5$, so that we attribute to both equilibria the same likelihood, conditional on their existence. The model's fit to the data is presented in Table A.1.

3.2 Global Solution Method

We solve the model using global solution methods. This is paramount to capture the nonlinear effects of financial crises on the macroeconomy and to allow for non-monotonic impacts of climate policy on the likelihood of financial crises. Specifically, we use time iteration with linear interpolation on a discretized state space. The model has two en-

[8]Formally, the loss parameter $\gamma_D = 0.0115$ in the period utility function can be backed out using the following relationship:

$$\mathbb{E}_{2025}\Big[u\Big((1-\gamma_T\cdot\Delta\mathcal{T}_t)C_t, \cdot, \Delta\mathcal{T}_t = 2.7°C; \gamma_D = 0\Big)\Big] = \mathbb{E}_{2025}\Big[\beta^{T+t}\Big(C_t, \cdot, \Delta\mathcal{T}_t = 2.7°C; \gamma_D > 0\Big)\Big],$$

where we set $t = 2100$ and all arguments entering household welfare are evaluated under the adverse scenario of maintaining current policies indefinitely and taking into account the non-linear dynamics of the model. We focus on a temperature increase by $2.7°C$ since the model-implied temperature increase by 2100 corresponds to $2.7°C$ without further climate policy, see Figure 1.

Table 1: Calibration

a) Conventional parameters		Value	Target / Source
Discount factor	β	0.9975	Risk free rate of 1.0% p.a.
Frisch labor elasticity	$1/\gamma^L$	0.75	Chetty et al. (2011)
Risk aversion	γ^C	1	Log utility for consumption
TFP level	A	0.407	Output normalization
Government spending	G	0.2	Govt. spending to output ratio of 20%
Capital share	α	0.33	Capital income share of 33%
Capital depreciation	δ	0.025	Depreciation rate of 10% p.a.
Price elasticity of demand	ϵ	10	Markup of 11%
Rotemberg adjustment costs	ρ^r	178	Calvo duration of 5 quarters
Investment cost intercept	a_0	-.008	Normalization of $\Gamma(I/K) = I$
Investment cost slope	a_1	0.530	Asset price normalized to 1
Investment cost curvature	a_2	0.25	Bernanke, Gertler, and Gilchrist (1999)
Target inflation	Π	1.005	Inflation target of 2%
MP response to inflation	κ_Π	2.0	Conventional value
MP response to output	κ_y	0.125	Conventional value
b) Climate parameters		**Value**	**Target / Source**
Clean technology cost slope	$b_{1,1990}$	0.075	In line with Nordhaus (2018)
Clean technology cost slope	$b_{1,2090}$	0	In line with Nordhaus (2018)
Clean technology cost curvature	b_2	1.6	In line with Nordhaus (2018)
Temperature response	Ψ_{CCR}	$0.58°C$	Fernandez-Villaverde, Gillingham, and Scheidegger (2025)
Population growth	g	0.0025	Growth rate 1% p.a.
GDP loss	τ_T	0.077	In line with Nath, Ramey, and Klenow (2024)
Temperature loss	γ_D	0.0115	In line with Nath, Ramey, and Klenow (2024)
c) Financial sector and shock parameters		**Value**	**Target / Source**
Target intermediation cost HH	γ^F	0.38	Share financial sector
Slope intermediation cost HH	ω^F	0.0424	Leverage multiple of 15
Mean risky technology	ψ	0.01	Rottner (2023)
Survival rate	ζ	0.885	Credit spread of 90bp
Persistence risk	ρ^ξ	0.96	Rottner (2023)
Std. dev. risk shock	σ^ξ	0.0031	Financial crisis probability = 2%
Sunspot shock probability	Υ	0.5	Normalization

dogenous state variables, total capital K_t and financial sector net worth N_t, and two exogenous states, the risk shock ϵ_t^ξ and sunspot shock ϵ_t^π.

Transition paths are solved by backward induction starting from the terminal long-run equilibrium with net zero emissions. While the initial change in the transition speed is an unexpected shock, we account for uncertainty along the transition path as agents are aware of the materialization of shocks. Our solution method allows us to solve the equilibrium for any imposed carbon tax path. We refer to Appendix A.2 for details on the numerical solution method.

4 Quantitative Analysis

In this section, we use our calibrated model to study the financial stability implications of climate policy, proceeding in three steps. First, we study the transition dynamics from the current climate policy trajectory onto a carbon tax path, which is consistent with emission reduction goals specified in the Paris Agreement. Second, we evaluate the net financial stability effect of such a climate policy shift. Third, we benchmark the welfare effects of "Climate Minsky Moments" against the real costs and benefits of the net zero transition.

4.1 Financial Stability Along the Transition to Net Zero

First, we evaluate the impact of carbon taxes on financial stability during the net zero transition. We consider different carbon tax paths $\{\tau_{t+i}^{c,j}\}_{i=0}^{\infty}$, where the second superscript j indicates the path. As a reference scenario, we construct the *current trajectory* of emissions, which is denoted by $\{\tau_{t+i}^{c,current}\}_{i=0}^{\infty}$ and keeps carbon taxes at their current global average of 5\$/tCO2. As our main policy experiment, we then consider an unanticipated shift from the low initial carbon tax to a stringent *Paris-aligned* climate policy path, which we denote by $\{\tau_{t+i}^{c,Paris}\}_{i=0}^{\infty}$ in the following. We interpret the year 2025 as the initial period T_0 in which the economy unexpectedly shifts onto the *Paris-aligned* path. Here, carbon taxes increase linearly until they reach their net-zero-consistent level $\tau_t^c = b_{1,2050}$ in 2050, such that all goods are produced using the emission-free technology. Emission taxes stay at the net-zero consistent level afterwards, and there is no uncertainty about the carbon tax path once it is revealed.

The dashed red lines in Figure 1 reflect the *current trajectory*, while the *Paris-aligned* path is indicated by solid blue lines. The top left panel shows the carbon tax paths, while the abatement cost slope declines irrespective of the climate policy path. The top right panel demonstrates that a 100% clean product share is achieved in 2050 under the *Paris-aligned* path. Due to technological change, net zero emissions are reached in 2090 even without climate policy tightening.

The middle panel of Figure 1 displays the implications of different carbon tax paths for carbon emissions, which are closely related to the clean product share η_t^*. At each point in time, emissions are substantially smaller under the *Paris-aligned* path. On the *current trajectory*, the global temperature increases by slightly more than $2.7°C$, which clearly misses the goals of the Paris Agreement. In contrast, the ambitious transition manages to keep global warming at a manageable level. Consistent with the prediction from similar DSGE models, such as van der Ploeg and Rezai (2020), and recent empirical evidence (Kaenzig, 2023), a faster transition induces a stronger output contraction and a larger wedge in the return on capital in the short run.

The next step is to evaluate the impact of these different transition paths on macroeconomic outcomes and financial stability, as measured by the crisis probability. An essential element in our analysis is that our framework is stochastic. To account for this, we simulate the economy along a given carbon tax path 100.000 times, where the risk shock and the sunspot shock are drawn randomly so that the volatility of idiosyncratic productivity shocks follows equation (12) and the sunspot shock selects the equilibrium if multiplicity

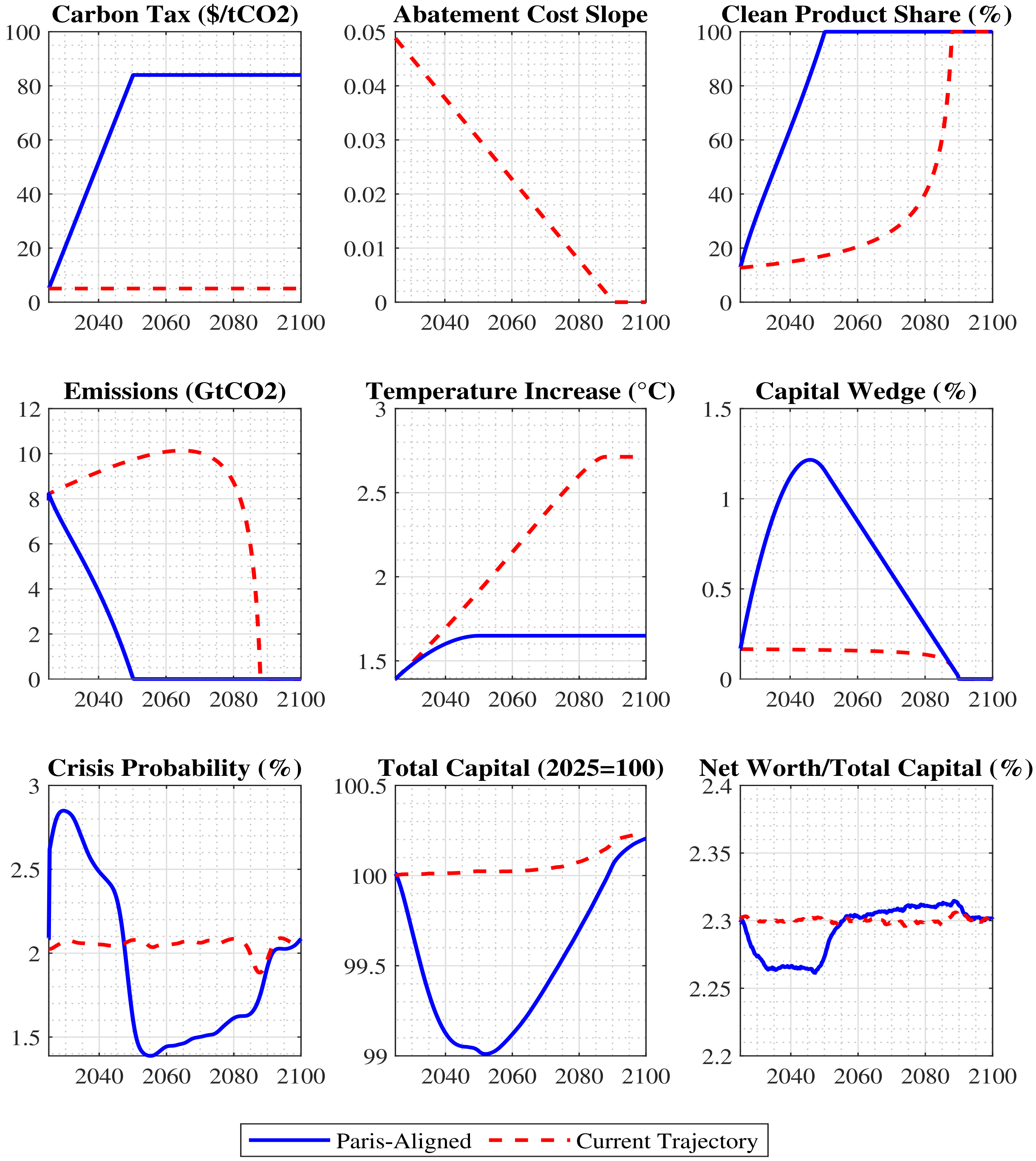

Figure 1: Transition path to net zero. Comparison of the Paris-aligned tax path and the current trajectory on selected variables. The capital return wedge, equation (2), combines carbon tax expenses and costs of operating emission-free technologies per unit of capital. The dynamics are shown for the years 2024Q4 until 2100Q4. The results are based on simulating the model 100,000 times with a burn-in period of 200 quarters. Crisis probabilities are annualized, and cubic spline smoothing is applied to remove sampling error for improved readability.

is possible. We then average over all economies to calculate the crisis probability and other macroeconomic variables.[9]

[9]To enhance the readability of our graphs, we apply a cubic spline filter to the crisis probability. Figure B.1 provides the raw figure. All macroeconomic moments and the welfare measures are always computed based on the unfiltered simulation output.

The bottom left panel of Figure 1 shows the crisis probability along the transition. It increases from around 2% on the initial path to around 2.7% at the beginning of the *Paris-aligned* transition. At this point in time, the economy experiences an unanticipated shock to the carbon tax path and, thus, a negative permanent shock to the marginal product of capital. This event puts deleveraging pressure on the financial system, which raises the run probability. A smaller adverse risk shock realization suffices to tighten intermediaries' leverage constraints by so much that they have to sell large amounts of capital. Formally, the partition of the state space supporting the run equilibrium under the depressed capital productivity suddenly increases. Notably, the crisis probability peaks several years into the transition and only slowly converges to its lower long-run level. Since capital and net worth are endogenous state variables, our model features a large degree of endogenous propagation. The bottom right panel shows that the intermediaries' net worth, normalized by the total capital stock, remains low for several years, consistent with the persistently elevated run probability. Appendix B.1 additionally displays impulse response functions and the evolution of several key variables during the transition, highlighting the outlined dynamics.

The bottom right panel of Figure 1 also reveals that the annualized crisis probability undershoots its long-run level of 2% to around 1.4% under the *Paris-aligned* transition path. As long as taxes and abatement costs are both high, the wedge in the return on capital is high, and there is less capital accumulation, as shown in the bottom middle panel. In times of stress, households therefore need to absorb fewer assets and are willing to pay a higher price for capital. This directly follows from their period utility function (8) and implies that banks can deleverage more easily in downturns, thereby reducing the run frequency. During these periods, intermediary net worth, normalized by total capital, is comparatively larger than in the initial and terminal equilibrium. Further details are provided in Appendix B.2.

Once the cost parameter reaches zero, the productivity losses from operating the clean technology vanish. Productivity is even slightly larger than in the initial equilibrium with high abatement costs and a 5\$/tCO2 carbon tax. This explains why capital rises toward the end of the simulation horizon and reaches a level slightly above its 2025 value, as firms can adopt the clean technology without incurring any abatement costs. The crisis frequency briefly drops once these productivity gains materialize and reverts to its initial value of 2% in the terminal equilibrium with costless green technologies, in which carbon taxation does not affect macroeconomic outcomes.

4.2 Net Financial Stability Effect

The next step is to evaluate the net financial stability effect of a carbon tax, which requires defining a suitable metric. Such a metric needs to take into account two crucial aspects. First, financial crises also occur in the absence of climate policy, so that an appropriate quantification of the threat of "Climate Minsky Moments" requires taking into account "Ordinary Minsky Moments" unrelated to ambitious carbon taxes. Second, since ambitious climate policy has a non-monotonic effect on the crisis probability along the transition, aggregating this effect over time is the key determinant of the net welfare effect.

To take these two aspects into account in a single metric, we define the *excess crisis probability (ExCP)*:

$$ExCP(\widetilde{\beta}) = \sum_{t=2025Q1}^{\infty} \widetilde{\beta}^t \left(\pi_t^{Paris} - \pi_t^{current}\right) , \tag{27}$$

where π_t^{Paris} is the probability of a financial crisis in period t on the tax path in line with the Paris Agreement and $\pi_t^{current}$ corresponds to crisis probability in period t on the *current trajectory*. By computing the net present value of the financial fragility induced by ambitious climate policy, the *excess crisis probability* bears conceptual resemblance to the social cost of carbon, which is a commonly used metric in climate economics.

As stressed by Nordhaus (2007) and Weitzman (2007), social discounting is a crucial element in the evaluation of climate policies, which we capture by the parameter $\widetilde{\beta}$. While we do not take a stand on the appropriate social discount rate, the possibility of discounting future financial stability gains via the social discount factor puts different weights on the elevated crisis probability at the outset versus the long-run stability gains. Graphically, the *ExCP* is represented by the area between the crisis probability under the *Paris-aligned* path and the *current trajectory*, see the left panel of Figure 1. When the social discount factor is set to $\widetilde{\beta} = 1$, the excess crisis probability is obtained simply by subtracting the two areas.

The left panel of Figure 2 shows the *ExCP* for varying annualized social discount factors. The results highlight that the net financial stability effect depends on the social discount rate. If the discount rate is higher than 1.5 p.a., the *ExCP* is positive due to the increased fragility at the beginning of the transition. The *Paris-aligned* transition has a negative financial stability effect in this case. For social discount rates below 1.5% p.a., there is even a net positive financial stability effect. Thus, the emergence of a trade-off between climate policy and financial stability depends on the patience of the policymaker.

4.3 The Welfare Relevance of "Climate Minsky Moments"

Having discussed the net financial stability of climate policy and its ambiguous sign, we now turn to the key normative question of our paper: how important are "Climate Minsky Moments" in the general welfare assessment of climate policies? Social welfare is based on households' period utility function:

$$W_t(\{\tau^{c,j}_{t+i}\}^{\infty}_{i=0}) = \sum_{i=0}^{\infty} \widetilde{\beta}^i \mathbb{E}_t \left[\frac{(C^j_{t+i})^{1-\gamma_C}}{1-\gamma_C} - \frac{(L^j_{t+i})^{1+\gamma_L}}{1+\gamma_L} - \mathcal{C}(K^{H,j}_{t+i}, K^j_{t+i}) - \mathcal{D}(\Delta\mathcal{T}^j_{t+i}) \right], \tag{28}$$

where $\widetilde{\beta}$ is the social discount factor and the superscript j emphasizes that all equilibrium objects and welfare depend on the imposed carbon tax path $\{\tau^{c,j}_{t+i}\}^{\infty}_{i=0}$. It should be stressed that all welfare results do not depend on including households' capital management cost as a non-standard element into the welfare function. Intuitively, the welfare losses of financial crises are almost entirely driven by the deep recession and consumption losses associated with runs, rather than with the direct cost of managing physical capital. We refer to Figure B.6 in Appendix B.4 for details.

The total welfare effect of the *Paris-aligned* policy relative to the *current trajectory* is formally defined as

$$\Delta W^{total}_t = W_t\big(\{\tau^{c,Paris}_{t+i}\}^{\infty}_{i=0}\big) - W_t\big(\{\tau^{c,current}_{t+i}\}^{\infty}_{i=0}\big)\,. \tag{29}$$

It is straightforward to decompose the total welfare effect into a temperature gain component, a cost component reflecting productivity losses from adopting and operating clean technologies, and the macroeconomic costs of financial crises. Formally, $\Delta W^{total}_t = \Delta W^{temp}_t + \Delta W^{cost}_t$, where the welfare gain from temperature damages is given by the difference in the damage component in equation (8) under the different tax paths:

$$\Delta W^{temp}_t = \sum_{i=0}^{\infty} (\widetilde{\beta})^i \mathbb{E}_t \left[\mathcal{D}(\Delta\mathcal{T}_{t+i}(\tau^{c,Paris}_{t+i})) - \mathcal{D}(\Delta\mathcal{T}_{t+i}(\tau^{c,current}_{t+i})) \right]\,. \tag{30}$$

The welfare costs from productivity losses and "Climate Minsky Moments" are given by

$$\Delta W^{cost}_t = \sum_{i=0}^{\infty} \widetilde{\beta}^i \mathbb{E}_t \left[\overline{u}(\tau^{c,Paris}_{t+i}) - \overline{u}(\tau^{c,current}_{t+i}) \right]\,. \tag{31}$$

Here, $\overline{u}(\{\tau^{c,j}_{t+i})$ collects all elements of the household utility function, equation (8), other than the losses from global warming, i.e., consumption, leisure, and capital management costs.

We isolate the welfare effect of "Climate Minsky Moments" from the overall costs of the net zero transition (31) by constructing a counterfactual that includes the *Paris-aligned* carbon tax path, but the run probabilities of the *current trajectory*. Formally, we subtract and add the utility from this counterfactual path $\overline{u}(\tau^{c,Paris}_{t+i})\big|_{\{\pi^{current}_{t+i}\}^{\infty}_{i=0}}$ to equation (31) to then decompose the welfare measure. The welfare costs of "Climate Minsky Moments" are the difference in utility that stems from the different run probabilities for the *Paris-*

aligned path and *current trajectory*. Climate policy itself is fixed at the *Paris-aligned* path. Formally, this can be written as:

$$\Delta W_t^{Minsky} = \sum_{i=0}^{\infty} (\widetilde{\beta})^i \mathbb{E}_t \left[\overline{u}(\tau_{t+i}^{c,Paris})\big|_{\{\pi_{t+i}^{Paris}\}_{i=0}^{\infty}} - \overline{u}(\tau_{t+i}^{c,Paris})\big|_{\{\pi_{t+i}^{current}\}_{i=0}^{\infty}} \right] . \tag{32}$$

Note that we define, for consistency, $\overline{u}(\tau_{t+i}^{c,Paris}) \equiv \overline{u}(\tau_{t+i}^{c,Paris})\big|_{\{\pi_{t+i}^{Paris}\}_{i=0}^{\infty}}$. This formulation highlights that the tax path and the run probabilities come both from the same *Paris-aligned* transition.

The welfare losses from productivity losses are then given by the difference between the two tax paths, in which the run probability is kept constant at the *current trajectory*:

$$\Delta W_t^{prod} = \sum_{i=0}^{\infty} (\widetilde{\beta})^i \mathbb{E}_t \left[\overline{u}(\tau_{t+i}^{c,Paris})\big|_{\{\pi_{t+i}^{current}\}_{i=0}^{\infty}} - \overline{u}(\tau_{t+i}^{c,current})\big|_{\{\pi_{t+i}^{current}\}_{i=0}^{\infty}} \right] . \tag{33}$$

It is worth noting that "Climate Minsky Moments" also affect the welfare losses from global warming. Financial crises are associated with deep and long-lasting recessions that somewhat mitigate global warming by depressing economic activity. As we demonstrate in Figure B.1, this effect is quantitatively negligible, so we base our welfare measure of "Climate Minsky Moments" solely on the cost component.

The left panel of Figure 2 displays the welfare effect of "Climate Minsky Moments" as blue bars in consumption equivalents. They generally have the inverse shape of the *ExCP*, as a larger crisis probability is associated with welfare losses. Excess crisis probability and welfare are zero and change signs for a social discount rate of around 1.5%. Hence, for a social discount rate close to the risk-free rate implied by our calibration, we obtain an approximate irrelevance result. Financial stability considerations have no effect on the total welfare impact of ambitious climate policies.

The right panel of Figure 2 shows that the threat of "Climate Minsky Moments" does not substantially affect the basic climate policy trade-off between productivity losses associated with stringent carbon taxes (red bars) and gains from a reduction of global surface temperatures (green bars), even for social discount rates different from 1.5%. Note that the figure does, in fact, feature the welfare effect of "Climate Minsky Moments" as blue bars, but they are barely visible due to their considerably smaller scale. The welfare gains from curbing global warming amount to approximately 11% in consumption equivalents for a social discount rate near zero, greatly exceeding the productivity losses from the clean technology (around 3%). As the social discount rate increases, temperature gains rapidly lose relevance. For social discount rates exceeding 2%, the net effect turns negative.

Taken together, our quantitative welfare analysis suggests that changed financial fragility provides net welfare gains if policymakers are very patient. For high social

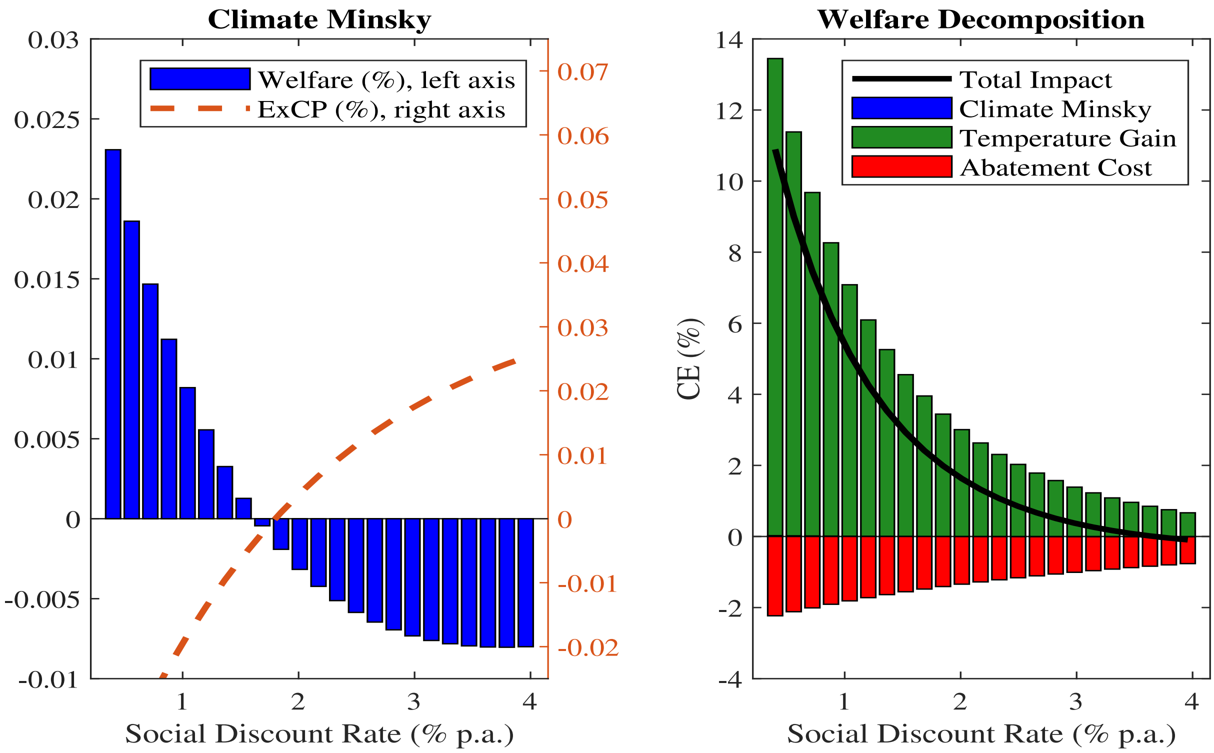

Figure 2: Welfare and financial stability effects. Left panel: Welfare effect of "Climate Minsky Moments" (left axis) and the excess crisis probability for varying social discount rates. Right panel: Decomposition of total welfare into contributions from "Climate Minsky Moments", temperature gains, and productivity losses. The results are based on 100,000 model simulations with a burn-in period of 200 quarters.

discount rates, they inflict a slight welfare loss that amounts to less than one percent of the total losses associated with the transition to emission-free technologies.

5 Extensions and Robustness

In this section, we demonstrate that the short-run financial fragility, the medium-run financial stability gain of "Climate Minsky Moments" and its relatively small welfare relevance compared to the real costs and benefits of the net zero transition are robust to variations in the climate block and to different climate policy specifications. Concerning climate damages, our baseline model assumes that the emission damages associated with global warming materialize through the utility function, which implies that there is no feedback to the incentives to accumulate capital. In Section 5.1, we relax this assumption and introduce physical climate damages. Temperature changes directly affect the aggregate production function and, in turn, the incentives to accumulate capital. In Section 5.2, we consider different climate policy specifications, such as abatement subsidies, emission performance standards, or the year at which net zero is reached.

Table 2 summarizes our results by comparing key metrics across the different extensions as well as modifications: the maximum difference between crisis probabilities

Table 2: Financial stability, climate, and welfare outcomes for various modifications

	Max Δ crisis pr. $\%^a$	Excess crisis pr. $\%^b$	Welfare Minsky CE $\%^c$	Welfare prod. CE $\%^d$	Welfare temp. CE $\%^e$	Temp. change $°C^f$
Baseline model	1.23	-0.01	0.003	-1.61	5.07	1.65
Physical damages	0.98	0.15	-0.08	Net Effect: 6.74 [g]		1.65
Higher damages	1.23	-0.01	0.003	-1.61	10.22	1.65
Clean tech. subsidies	0.78	-0.02	0.008	-1.78	5.69	1.58
Env. perf. standard	1.35	0	-0.004	-1.50	5.06	1.65
Net zero in 2055	0.86	-0.01	0.004	-1.46	4.77	1.69
Back-loaded transition	1.30	-0.01	-0.003	-1.48	4.66	1.70

[a] Maximum difference between (annualized) financial crisis probability $\pi_t^{Paris} - \pi_t^{current}$.
[b] Excess crisis probability (ExCP) computed for $\widetilde{\beta} = 0.9975$.
[c] Welfare effect of Climate Minsky Moments in consumption equivalents with $\widetilde{\beta} = 0.9975$.
[d] Welfare effect due to productivity losses in consumption equivalents with $\widetilde{\beta} = 0.9975$.
[e] Welfare effect due to temperature in consumption equivalents with $\widetilde{\beta} = 0.9975$.
[f] Temperature change from 1850 to 2100 under the Paris-aligned transition, expressed in degrees Celsius.
[g] The net welfare effect for physical risk is reported since no exact welfare decomposition is possible in this model variant.

between the Paris-aligned and the current trajectory (column 1), the excess crisis probability (column 3), welfare effect of Climate Minsky Moments (column 4), the welfare costs of productivity losses (column 5) and the welfare gain due to emission reduction (column 6), which is directly related to the temperature change (column 7).

5.1 Climate Damages

In our baseline model, damages associated with global warming affect welfare directly through the household utility function. Capturing the welfare benefit of ambitious climate policy in this way is quite common in the literature (Acemoglu et al., 2012; Barrage, 2019), and it is possible to exactly decompose the welfare effect of climate policy into productivity losses, temperature gains, and financial stability implications. However, the climate economics literature also considers models where global warming negatively affects aggregate TFP.

In this section, we show that our results are robust to including climate change damages directly in the aggregate production function. This alternative formulation is given by

$$Y_t = \exp(-\gamma_T \Delta \mathcal{T}_t) A K_{t-1}^{\alpha} L_t^{1-\alpha} \ . \tag{34}$$

Consequently, we drop the utility losses from temperature increases incurred by households. The parameter γ_T is informed directly from recent empirical work by Nath, Ramey, and Klenow (2024). Note that in these extensions, cumulated emissions become an additional endogenous state variable, which we take into account in the numerical solution method.

Since emissions and hence climate damages are relatively lower on the *Paris-aligned* carbon tax path than on the *current trajectory*, the economy accumulates relatively more capital. Consequently, the associated medium-run financial stability gains of the *Paris-aligned* climate policy are less pronounced. Therefore, the excess crisis probability is positive for all relevant social discount rates. Table 2 demonstrates that this also implies a more pronounced welfare loss of “Climate Minsky Moments”. Besides the computational costs of adding an endogenous state variable to the model, the welfare decomposition into a temperature and an abatement cost component is not possible with physical damages. Therefore, we can only report the net welfare effect. Importantly, as in the baseline model, the financial stability implications are a negligible component of total welfare, as shown in Table 2.

Losses from Temperature Increases Another dimension that is important for welfare is how global temperature changes translate into losses to households. Specifically, we consider the upper bound estimated by Nath, Ramey, and Klenow (2024) to back out a large but plausible damage parameter γ_D. Their upper bound is a GDP loss of 3.2%, which is close to the estimate by Bilal and Rossi-Hansberg (2023). This estimate directly maps into a higher damage parameter $\gamma_D = 0.0147$ in the household utility function. While this re-parameterization does not affect global temperature increases under different climate policies, but merely the utility loss they inflict, this robustness check is conceptually similar to modifying the effect of cumulated emissions on temperature Ψ_{CCR}, but keeping γ_D constant. As Table 2 highlights, raising the damage parameter does not affect the welfare effect of “Climate Minsky Moments”, but merely increases the relative benefit of climate policy through temperature gains. Recall that temperature losses enter the household utility function in an additive fashion, such that the effect of different climate policy trajectories on the macroeconomy and financial stability is independent of the parameter γ_D.

5.2 Climate Policy Variations

This section explores the robustness of our results with respect to the specifications of climate policy and parameter choices in the climate block. Our analysis demonstrates the robustness of our result about the welfare (ir)relevance of “Climate Minsky Moments”. In all these modifications, the crisis probability increases initially, even though the peak effect varies slightly, while there are substantial financial stability gains during later stages of the transition. Our finding that the welfare effect of “Climate Minsky Moments” is clearly second-order also holds across all specifications. While there are some substantial changes in the welfare effects from curbing climate change in the different modifications, the relative ranking concerning financial stability is clearly untouched.

Clean Technology Subsidies We first examine the role of fiscal policy and allow for the possibility of clean technology subsidies through the lens of our model. So far, we have assumed that carbon tax revenues are rebated to households in a lump sum fashion. We demonstrate that our key results are robust to assuming that the carbon tax revenue is used to subsidize firms' adoption of clean technologies.[10] Specifically, firms receive a flat subsidy per abated unit of emissions $\eta_t Y_t$, which is financed with carbon tax revenues. The minimization problem, shown in Appendix B.3, implies that the share of clean firms is larger than in the baseline model for any given carbon tax. Furthermore, the associated wedge in the return on capital is smaller than under the assumption of tax rebates to households.

Such a policy substantially impacts the transition dynamics for the *Paris-aligned* tax path. The crisis probability increases very slowly over the first ten years of the transition. The subsidy cushions intermediary net worth against rapid drops in the return on their assets, which entails a financial stability gain at the outset of the transition. At the same time, this makes the downward adjustment of capital more sluggish. Therefore, the economy operates a larger capital stock well into the transition compared to the case without subsidies. Once tax revenues approach zero, the per-unit subsidy becomes small, and asset returns decline. Intermediaries face increasing pressure to sell capital, which is still costly for households to absorb. The crisis probability, thus, remains above the *current trajectory* throughout the later stages of the transition. The long-run crisis probability is unchanged, as there is no tax revenue once net zero is reached.

Taken together, this results in an *excess crisis probability* that is above the baseline scenario. From a welfare perspective, the benefits of reduced temperature increases outweigh their costs, as Table 2 highlights. Our analysis suggests that subsidies - a popular alternative policy instrument to accelerate the net zero transition - conflict to some extent with financial stability objectives. However, the climate policy trade-off is heavily solved in favor of subsidies due to their large positive effect on the adoption of clean technologies and the associated emission reduction.

[10] While climate policy has usually focused on carbon pricing and R&D subsidies for emission-free technologies, policymakers have recently started to subsidize the adoption of low-emission technologies at a large-scale. Examples include the "Inflation Reduction Act" in the US and the European Commission's "Green New Deal". For an analysis of different fiscal policy regimes in the context of transition risk, see Carattini et al. (2024). Jondeau et al. (2025) show in an estimated E-DSGE model that abatement subsidies financed by emission tax revenues promote firm entry into the abatement sector and thereby reduce the cost of switching to clean technologies.

Emission Performance Standards Next, we explore the implications of implementing climate policy through emission performance standards rather than pricing emissions. Such climate policies have strong practical relevance, as China, for example, operates a so-called "Tradable Performance Standard" that specifies target emission intensities. We map such a policy into our model by assuming that the regulator mandates the use of clean technologies for the share $\widetilde{\eta}_t$ of all goods varieties. Naturally, firms will choose the technologies that are cheapest to operate in a clean fashion. Importantly, firms still pay the abatement cost, which in turn reduces the payoff from their investment and thereby lowers aggregate capital accumulation. However, firms do not have to pay emission taxes on the share of goods that are produced with the dirty technology. The modified return on capital wedge is simply given by $\widetilde{B}_t = \frac{b_{1,t}}{b_2+1}(\widetilde{\eta}_t)^{b_2+1}$.

As Table 2 demonstrates, such a policy has very similar implications for the long-run crisis probability and temperature changes. This is not surprising since the total abatement cost incurred by firms is identical to the carbon tax, once net zero is reached. However, such a policy implies a different time pattern of the return on capital wedge $\widetilde{B}_t$. At the onset of the transition and on the *current trajectory*, the wedge is very low due to the low initial abatement cost and because firms do not have to pay taxes on their initially high emissions. Hence, productivity, capital accumulation, and financial leverage remain high. However, the shift on the *Paris-aligned* path induces a relatively stronger increase in the return on capital wedge $\widetilde{B}_t$, when compared to carbon taxes, where firms incur the same abatement cost but experience smaller carbon tax bills. Hence, financial fragility spikes to a slightly higher level and remains elevated towards the end of the transition. Consequently, the welfare losses of "Climate Minsky Moments" are larger than in the baseline model.

Slowing Down the Transition While the Paris Agreement envisions a path to net zero by 2050, we also consider a tax path that reaches net zero five years later (T_{max} = 2055), with the corresponding results reported in the second-to-last row in Table A.1. Such a policy clearly has negative implications for global warming. The difference amounts to around 0.07° by the end of the century. At the same time, the initial increase in financial instability is less pronounced. However, the *excess crisis probability* in total is rather unaffected, as the medium-run financial stability gains associated with less capital accumulation are also less pronounced. While the welfare gains from curbing global warming are much smaller than in the baseline specification, the second-order relevance of "Climate Minsky Moments" is unchanged.

Back-Loaded Transition To take advantage of the medium-run financial stability gains without temporarily elevated crisis probabilities, it is reasonable to ask whether temporary delays of the transition yield superior net financial stability outcomes along the net zero transition. To ensure a fair comparison to the baseline tax path, we do not change the target year for net zero, such that there are more aggressive carbon tax hikes at a later stage. This introduces a convexity into the carbon tax path and is sometimes referred to as a "disorderly transition." The results are summarized in the last row of Table A.1. The back-loaded transition does not yield an improved financial stability outcome. Instead, it spreads the elevated crisis probabilities over time, but with a lower peak crisis probability in the short run. Again, the welfare gains from curbing global warming are less pronounced, since emissions remain high in the initial periods with a high abatement cost parameter and high emissions. However, the ExCP and the net welfare effects are quite similar to the linear tax path.

6 Macroprudential and Monetary Policy

Our analysis has important implications for the interactions between climate policy and the conduct of macroprudential and monetary policies. We first show that the financial cycle is a crucial driver of the financial stability effects associated with stringent climate policies. We then discuss how monetary policy can help alleviate the negative financial stability effects of the transition. This turns out to depend critically on the shape of the climate policy path. Again, the endogeneity of the financial sector is key.

6.1 Macroprudential Policy: The Role of the Financial Cycle

The possibility of a "Climate Minsky Moment" depends jointly on the (exogenous) climate policy stance and the (endogenous) loss-absorbing capacity of the financial sector, i.e., its net worth. We illustrate how the loss-absorbing capacity shapes the financial stability implications of climate policy action. We focus on the same *Paris-aligned* transition path as before, but condition our simulation on specific realizations of the risk shock, which endogenously affects the financial sector's loss-absorbing capacity.

First, we consider a sequence of negative (one standard deviation) risk shock realizations in two quarters prior to the climate policy shift. This period of low volatility induces high leverage and a credit boom, capturing a volatility paradox in the spirit of Brunnermeier and Sannikov (2014). Once the surprise climate policy shift arrives, the financial sector is highly leveraged and lacks loss-absorbing capacity. As a consequence, the annualized crisis probability spikes to slightly more than 5% and stays elevated way into the transition. The dashed red line in the right panel of Figure 3 shows the run probability in the high-risk case.

The opposing path, i.e., a sequence of positive (one standard deviation) risk shock realizations, forces the financial sector to deleverage prior to the climate policy shift. When the shift arrives, the financial sector is much better equipped to accommodate

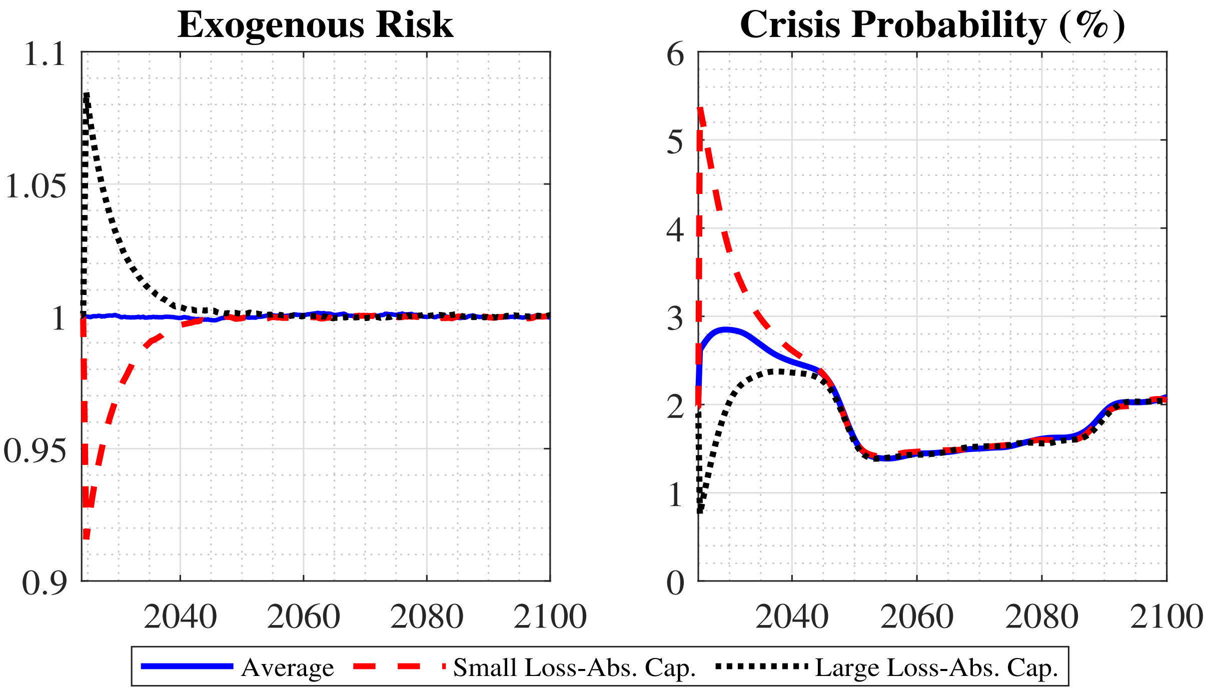

Figure 3: Role of the financial cycle. Impact of loss-absorbing capacity on the crisis probability (right panel). The scenarios with a small (large) loss-absorbing are obtained by imposing an additional risk shock realization of minus (plus) one standard deviation relative to the baseline scenario in the two quarters preceding the shift towards the Paris-aligned path, as shown in the left panel. TThe results are based on simulating the model 100,000 times with a burn-in period of 200 quarters. Crisis probabilities are annualized, and cubic spline smoothing is applied to remove sampling error for improved readability.

the sudden productivity loss without selling securities at a fire sale price. The crisis probability, displayed as the dotted green line in the right panel of Figure 3, declines substantially. The crisis probability remains persistently low, as the capital accumulation channel of climate policy dominates in the medium-run. Appendix B.4 contains additional results for the net financial stability effect and welfare.

This analysis outlines that a careful design of the transition to net zero should take vulnerabilities in the financial system into account. The threat of a "Climate Minsky Moment" after the climate policy shift depends to a substantial degree on the loss-absorbing capacity of the financial sector. The key implication for macroprudential policymakers is to ensure that the loss-absorbing capacity is large in order to facilitate a smooth transition to net zero. Our results point towards welfare gains from coordinating macroprudential and climate policies.

6.2 Monetary Policy

The possibility of "Climate Minsky Moments" also has implications for monetary policy within central banks' financial stability mandates. Monetary policy can accommodate the transition by initially reducing nominal interest rates. This affects the probability of financial crises in two ambiguous ways: by increasing asset valuations, it reduces

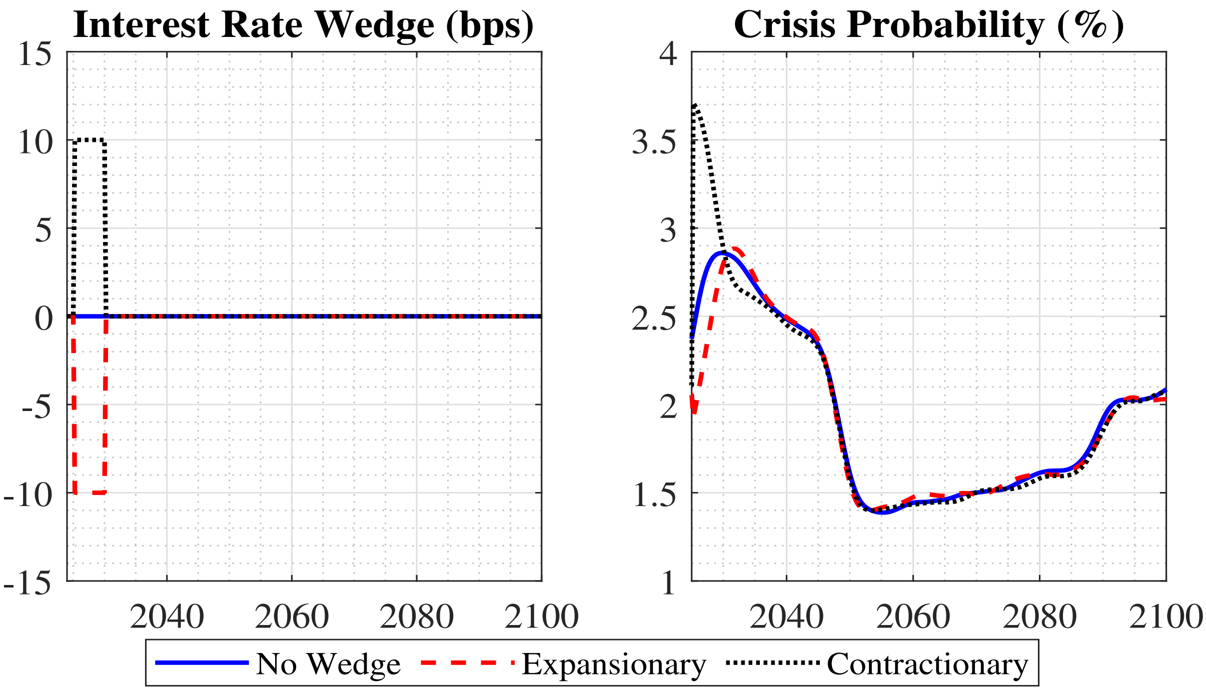

Figure 4: Role of the monetary policy stance. Impact of alternative monetary stances (left panel) on the crisis probability (right panel). The expansionary (contractionary) scenario assumes a negative (positive) wedge of 10 bps in each period in the first 5 years after the switch to the Paris-aligned path. The results are based on simulating the model 100,000 times with a burn-in period of 200 quarters. Crisis probabilities are annualized, and cubic spline smoothing is applied to remove sampling error for improved readability.

the partition of the state space supporting runs. On the other hand, this incentivizes intermediaries to increase their leverage, with negative effects on loss-absorbing capacities and elevated crisis probabilities in the future. Containing surges in the financial crisis probability by expansionary monetary policy has beneficial short-run effects, which come at a medium-run cost.

Alternatively, monetary policy can lean against the transition. This is a particularly relevant case if high carbon taxes generate inflationary pressures (Kaenzig, 2023). Here, an increase in policy rates at the onset of the transition initially amplifies run probabilities by further depressing asset prices. At the same time, this increases the loss-absorbing capacity going forward. In both cases, the net effect depends on deleveraging incentives that carbon taxes exert on intermediaries and social discounting.

We study how monetary policy affects the likelihood of "Climate Minsky Moments" when it accommodates or leans against the transition. Holding the climate policy path fixed, we add a linear forward guidance term i_t^{wedge} to the monetary policy rule (24):

$$R_t^I = \max\left\{ R^I \left(\frac{\Pi_t}{\Pi}\right)^{\kappa_\Pi} \left(\frac{MC_t}{MC}\right)^{\kappa_y} + i_t^{wedge}, 1 \right\} .$$

The (annualized) wedge is set to 10bps (contractionary) or -10bps (expansionary) for every quarter in the first five years after the onset of the transition and to zero afterward, see the left panel of Figure 4. The right panel of Figure 4 shows that the crisis probability drops to around 1.8% (dashed red line) for expansionary monetary policy and then overshoots slightly before reverting back to its old path. The opposite pattern emerges for a contractionary forward guidance term. Our results imply that monetary policy should not "lean against the transition" to avoid increased financial instability. This is particularly relevant as the impact of the wedge is asymmetric. The contractionary stance has a relatively larger absolute effect on the crisis probability, which rises by one percentage point more than in the baseline case without wedges. An expansionary stance, instead, only lowers the crisis probability by approximately half a percentage point.

Does such a monetary policy adjustment mitigate the welfare effects of Climate Minsky Moments in a quantitatively meaningful way? To ensure a fair comparison between *Paris-aligned* transition paths and the *current trajectory*, we also solve and simulate the *current trajectory* under the modified monetary policy rule. As shown in Appendix B.4, contractionary monetary policy raises the excess crisis probability and induces a larger welfare loss of "Climate Minsky Moments". In particular, the social discount rate above which "Climate Minsky Moments" reduce welfare is around one percentage point lower if monetary policy is contractionary.

7 Conclusion

In this paper, we have shown that climate policy has non-trivial effects on financial stability. We propose, solve, and calibrate a nonlinear quantitative macroeconomic model with carbon taxes and endogenous financial crises and derive three main results. First, asset stranding decreases financial stability in the short-run when the economy moves unexpectedly onto an ambitious carbon tax path. In response to such a negative shock to the return on their assets, financial intermediaries face deleveraging pressure, which induces them to sell assets quickly, potentially at fire sale prices. This makes a systemic financial crisis more likely. Second, climate policy is not detrimental to financial stability in the medium run, since it depresses capital holdings, such that households need to absorb fewer assets from the financial sector in an economic downturn. This softens the asset price drop in a downturn and makes systemic financial crises less likely. Third, climate policy does not affect financial fragility in the long run, when polluting and emission-free technologies are equally productive.

We show that the net effect of these opposing forces on financial stability and welfare crucially depends on the social discount rate. For a sufficiently patient policymaker, there is no trade-off between achieving climate policy goals and maintaining financial stability, since future stability gains receive a large welfare weight. If the social discount rate is high, a Paris-aligned transition implies welfare losses due to "Climate Minsky Moments". However, we demonstrate that the welfare losses associated with "Climate Minsky Moments" are of second-order importance compared to the welfare cost of adopting clean technologies. This result is robust to a large set of reasonable modifications, such as physical climate damages and variations in the specification of climate policy. Taken together, our results suggest that financial stability concerns are not a valid reason to delay ambitious climate policy action.

Throughout our analysis, we focus on the effects of climate change and climate policy on productivity *levels* and abstract from potential higher-order effects. Specifically, there is no *transition uncertainty*, i.e., uncertainty about carbon taxes, which have a theoretically ambiguous financial stability effect. While this represents an additional source of macroeconomic uncertainty, it also increases the incentives to invest in clean technologies (Fried, Novan, and Peterman, 2022). This might further raise financial fragility in the short run, but might also improve the medium-run financial stability gains. Furthermore, our framework abstracts from *acute* physical climate risks, such as more frequent natural disasters, which might become more frequent as global temperatures increase. Similar to transition uncertainty, acute physical risk has an ambiguous financial stability effect since it, on one hand, induces additional asset price volatility but also lowers the incentives to accumulate physical capital on the other hand. We leave an assessment of these higher-order effects for future research.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) to improve the language and readability of selected passages. The authors reviewed and edited all output and take full responsibility for the content of the article.

References

Abad, J., D. Martinez-Miera, and J. Suarez (2024). "A Macroeconomic Model of Banks Systemic Risk Taking." Working Paper.

Acemoglu, D., P. Aghion, L. Bursztyn, and D. Hemous (2012). "The Environment and Directed Technical Change." *American Economic Review* 102(1), 131–166.

Acemoglu, D., and F. Zilibotti (1997). "Was Prometheus Unbound by Chance? Risk, Diversification, and Growth." *Journal of Political Economy* 105(4).

Acharya, V. et al. (2023). "Climate Stress Testing." *Annual Review of Financial Economics* 15(1), 291–326.

Adrian, T., and H. S. Shin (2014). "Procyclical Leverage and Value-at-Risk." *Review of Financial Studies* 27(2), 373–403.

Airaudo, F., E. Pappa, and H. Seoane (2026). "The Green Metamorphosis of a Small Open Economy." *Journal of Money, Credit and Banking*.

Amador, M., and J. Bianchi (2024). "Bank Runs, Fragility, and Credit Easing." *American Economic Review* 114(7), 2073–2110.

Annicchiarico, B., M. Carli, and F. Diluiso (2026). "Climate Policies, Macroprudential Regulation, and the Welfare Cost of Business Cycles." Journal of Money, Credit and Banking.

Arlt, S., T. Berg, X. Hut, and D. Streitz (2025). "A Biodiversity Stress Test of the Financial System." Working Paper.

Barnett, M. (2023). "A Run on Fossil Fuel? Climate Change and Transition Risk." Working Paper.

Barrage, L. (2019). "Optimal Dynamic Carbon Taxes in a Climate–Economy Model with Distortionary Fiscal Policy." *Review of Economic Studies*.

Begenau, J., and T. Landvoigt (2022). "Financial regulation in a quantitative model of the modern banking system." *Review of Economic Studies* 89(4), 1748–1784.

Bernanke, B., M. Gertler, and S. Gilchrist (1999). "The Financial Accelerator in a Quantitative Business Cycle Framework." In: *Handbook of Macroeconomics*. Elsevier, 1341–1393.

Berthold, B., A. Cesa-Bianchi, F. D. Pace, and A. Haberis (2023). "The Heterogeneous Effects of Carbon Pricing: Micro and Macro Evidence." CEPR Discussion Paper 18312.

Bilal, A., and D. Kaenzig (2026). "The Macroeconomic Impact of Climate Change: Global vs. Local Temperature." *Quarterly Journal of Economics* 141, 889–944.

Bilal, A., and E. Rossi-Hansberg (2023). "Anticipating Climate Change Across the United States." Working Paper.

Blackburn, K., and V. Hung (1998). "A Theory of Growth, Financial Development and Trade." *Economica* 65, 107–124.

Blanco, A., C. Boar, C. Jones, and V. Midrigan (2024). "Non-linear Inflation Dynamics in Menu Cost Economies." Working Paper.

Boissay, F., F. Collard, and F. Smets (2016). "Booms and Banking Crises." *Journal of Political Economy* 124(2), 489–538.

Brunetti, C. et al. (2022). "Climate-related Financial Stability Risks for the United States: Methods and Applications." Working Paper.

Brunnermeier, M., and Y. Sannikov (2014). "A Macroeconomic Model with a Financial Sector." *American Economic Review* 104(2), 379–421.

Caggese, A., A. Chiavari, S. Singh Goraya, and C. Villegas-Sanchez (2024). "Climate Change, Firms, and Aggregate Productivity." Working Paper.

Campiglio, E., S. Dietz, and F. Venmans (2023). "Optimal Climate Policy as If the Transition Matters." Working Paper.

Carattini, S., G. Heutel, G. Melkadze, and I. Mourelon (2024). "Transition Risk: Sources and Policy Responses." Working Paper.

Carattini, S., G. Melkadze, and G. Heutel (2023). "Climate Policy, Financial Frictions, and Transition Risk." *Review of Economic Dynamics* 51, 778–794.

Carney, M. (2016). "Resolving the Climate Paradox." The Arthur Burns Memorial Lecture, 22 September.

Casey, G. (2024). "Energy Efficiency and Directed Technical Change: Implications for Climate Change Mitigation." *Review of Economic Studies* 91(1), 192–228.

Chabot, M., and J.-L. Bertrand (2023). "Climate risks and financial stability: Evidence from the European financial system." *Journal of Financial Stability* 69(101190).

Chetty, R., A. Guren, D. Manoli, and A. Weber (2011). "Are Micro and Macro Labor Supply Elasticities Consistent? A Review of Evidence on the Intensive and Extensive Margins." *American Economic Review* 101(3), 471–475.

Christiano, L., R. Motto, and M. Rostagno (2014). "Risk Shocks." *American Economic Review* 104(1), 27–65.

Cole, H., and T. Kehoe (2000). "Self-fulfilling Debt Crises." *Review of Economic Studies* 67(1), 91–116.

Diamond, D. (1984). "Financial Intermediation and Delegated Monitoring." *Review of Economic Studies* 51(3), 393–414.

Diamond, D., and P. Dybvig (1983). "Bank Runs, Deposit Insurance, and Liquidity." *Journal of Political Economy* 91(3), 401–419.

Dietrich, A., L. Leitenbacher, and G. Mueller (2026). "Consumer Durables, Monetary Policy, and the Green Transition." *European Economic Review* 182.

Diluiso, F., B. Annicchiarico, M. Kalkuhl, and J. Minx (2021). "Climate Actions and Macro-Financial Stability: The Role of Central Banks." *Journal of Environmental Economics and Management* 110, 102548.

Döttling, R., and M. Rola-Janicka (2025). “Too Levered for Pigou? A Model of Environmental and Financial Regulation.” *Journal of Financial Economics* 172(104105).

Elenev, V., T. Landvoigt, and S. Van Nieuwerburgh (2021). “A macroeconomic model with financially constrained producers and intermediaries.” *Econometrica* 89(3), 1361–1418.

Fernandez-Villaverde, J., K. Gillingham, and S. Scheidegger (2025). “Climate Change through the Lens of Macroeconomic Modeling.” *Annual Review of Economics* 17, 125–50.

Ferrari, A., and V. Nispi Landi (2023). “Toward a Green Economy: the Role of Central Bank’s Asset Purchases.” *International Journal of Central Banking* 19(5), 287–340.

Fornaro, L., V. Guerreri, and L. Reichlin (2024). “Monetary Policy for the Green Transition.” Working Paper.

Frankovic, I., and B. Kolb (2024). “The Role of Emission Disclosure for the Low-Carbon Transition.” *European Economic Review* 167, 104792.

Fried, S. (2018). “Climate Policy and Innovation: A Quantitative Macroeconomic Analysis.” *American Economic Journal: Macroeconomics* 10(1), 90–118.

Fried, S., K. Novan, and W. B. Peterman (2022). “Climate Policy Transition Risk and the Macroeconomy.” *European Economic Review* 147(104174).

Gertler, M., and N. Kiyotaki (2015). “Banking, Liquidity, and Bank Runs in an Infinite Horizon Economy.” *American Economic Review* 105(7), 2011–2043.

Gertler, M., N. Kiyotaki, and A. Prestipino (2020). “A Macroeconomic Model with Financial Panics.” *Review of Economic Studies* 87(1), 240–288.

Giovanardi, F., and M. Kaldorf (2026). “Climate Change and the Macroeconomics of Bank Capital Regulation.” *Journal of Monetary Economics* 159(103907).

Giovanardi, F., M. Kaldorf, L. Radke, and F. Wicknig (2023). “The Preferential Treatment of Green Bonds.” *Review of Economic Dynamics* 51, 657–676.

Golosov, M., J. Hassler, P. Krusell, and A. Tsyvinski (2014). “Optimal Taxes on Fossil Fuel in General Equilibrium.” *Econometrica* 82(1), 41–88.

Hassler, J., P. Krusell, and C. Olovsson (2021). “Directed Technical Change as a Response to Natural Resource Scarcity.” *Journal of Political Economy* 129(11), 3039–3072.

Heutel, G. (2012). “How Should Environmental Policy Respond to Business Cycles? Optimal Policy under Persistent Productivity Shocks.” *Review of Economic Dynamics* 15(2), 244–264.

Ikeda, D., and H. Matsumoto (2021). “Procyclical Leverage and Crisis Probability in a Macroeconomic Model of Bank Runs.” Working Paper.

IMF (2022). “United Kingdom: Financial Sector Assessment Program-Systemic Stress, and Climate-Related Financial Risks: Implications for Balance Sheet Resilience.” Technical Report.

Jermann, U., and V. Quadrini (2012). "Macroeconomic Effects of Financial Shocks." *American Economic Review* 102(1), 238–271.

Jondeau, E, G Levieuge, J Sahuc, and G Vermandel (2025). "Environmental Subsidies to Mitigate Net-Zero Transition Costs." *American Economic Journal: Macroeconomics, Forthcoming.*

Jondeau, E., B. Mojon, and C. Monnet (2021). "Greening (Runnable) Brown Assets with a Liquidity Backstop." BIS Working Paper 929.

Jordà, Ò., M. Schularick, and A. Taylor (2017). "Macrofinancial History and the New Business Cycle Facts." *NBER Macroeconomics Annual* 31(1), 213–263.

Jung, H., J. Santos, and L. Seltzer (2024). "U.S. Banks' Exposures to Climate Transition Risks." Federal Reserve Bank of New York Staff Reports, no. 1058.

Kaenzig, D. (2023). "The Unequal Consequences of Carbon Pricing." NBER Working Paper 31221.

Levine, R. (1997). "Financial Development and Economic Growth: vi6ws and Agenda." *Journal of Economic Literature* 35, 668–726.

Minsky, H. (1977). "A Theory of Systemic Fragility." In: *Financial Crises: Institutions and Markets in a Fragile Environment*. Edited by E Altman and A Sametz, 138–152.

Moreira, A., and A. Savov (2017). "The Macroeconomics of Shadow Banking." *Journal of Finance* 72(6), 2381–2432.

Nakov, A., and C. Thomas (2026). "Climate-Conscious Monetary Policy." *Journal of Political Economy: Macroeconomics* 4(2).

Nath, I., V. Ramey, and P. Klenow (2024). "How Much Will Global Warming Cool Global Growth?" NBER Working Paper 32761.

Nordhaus, W. (2007). "A Review of the Stern Review on the Economics of Climate Change." *Journal of Economic Literature* 45(3), 686–702.

——— (2008). *A Question of Balance: Weighing the Options on Global Warming Policies.* Yale University Press.

——— (2018). "Evolution of modeling of the economics of global warming: changes in the DICE model, 1992–2017." *Climatic Change* 148, 623–640.

Nuño, G., and C. Thomas (2017). "Bank Leverage Cycles." *American Economic Journal: Macroeconomics* 9(2), 32–72.

Oehmke, M., and M. Opp (2025). "Green Capital Requirements." *Journal of Finance, Forthcoming.*

Pagano, M. (1993). "Financial Markets and Growth: An Overview." *European Economic Review* 37, 613–622.

Richter, A., N. Throckmorton, and T. Walker (2014). "Accuracy, Speed and Robustness of Policy Function Iteration." *Computational Economics* 44, 445–476.

Robinson, J. (1952). "The Generalization of the General Theory." In: *The Rate of Interest, and Other Essays*. London: Macmillan, 67–142.

Rottner, M. (2023). "Financial Crises and Shadow Banks: A Quantitative Analysis." *Journal of Monetary Economics* 139, 74–92.
Sahuc, J.-G., F. Smets, and G. Vermandel (2024). "The New Keynesian Climate Model." Working Paper.
Saint-Paul, G. (1992). "Technological Choice, Financial Markets and Economic Development." *European Economic Review* 36(4), 763–781.
Schularick, M., and A. Taylor (2012). "Credit booms gone bust: Monetary policy, leverage cycles, and financial crises, 1870-2008." *American Economic Review* 102(2), 1029–61.
Stokey, N. (1988). "Are There Limits to Growth?" *International Economic Review* 39(1), 1–31.
Sussman, O. (1993). "A Theory of Financial Development." In: *Finance and development: issues and experience*. Ed. by A. Giovannini. Cambridge University Press, 29–64.
Van der Ploeg, F., and A. Rezai (2020). "Stranded Assets in the Transition to a Carbon-Free Economy." *Annual Review of Resource Economics* 12(1), 281–298.
Weitzman, M. (2007). "A Review of the Stern Review on the Economics of Climate Change." *Journal of Economic Literature* 45(3), 703–724.

A Model Appendix

A.1 Financial Intermediaries

In this section, we outline the maximization problem of financial intermediaries, which follows Rottner (2023).[11] The financial intermediary j maximizes its franchise value V_t^j subject to an incentive constraint and a participation constraint. The incentive constraint ensures that the intermediary only invests in the good technology. The participation constraint ensures that the return on deposits is sufficient that households provide deposits to the intermediary. The maximization problem reads as follows:

$$V_t^j(N_t^j) = \max_{K_t^{Bj}, \overline{D}_t} (1-\pi_t^j)\beta E_t^N \Lambda_{t,t+1}\left[\theta V_{t+1}^j\left(N_{t+1}^j\right) + (1-\theta)(R_{t+1}^K Q_t K_t^{Bj} - \overline{D}_t^j \Pi_{t+1}^{-1})\right], \tag{A.1}$$

$$\text{s.t.}\quad (1-\pi_t^j)E_t^N\left[\Lambda_{t,t+1}\theta V_{t+1}\left(N_{t+1}^j\right) + (1-\theta)\Big(1-\frac{\bar{b}_t^j}{R_{t+1}^K\Pi_{t+1}}\Big)R_{t+1}^K Q_t K_t^{Bj}\right] \geq \tag{A.2}$$

$$\beta\Lambda_{t,t+1}E_t\left[\Lambda_{t,t+1}\int_{\frac{\bar{b}_t^j}{R_{t+1}^K\Pi_{t+1}}}^{\infty}\theta V_{t+1}\left(N_{t+1}^j\right) + (1-\theta)\Big(\omega - \frac{\bar{b}_t^j}{R_{t+1}^K\Pi_{t+1}}\Big)R_{t+1}^K Q_t K_t^{Bj} d\tilde{F}_{t+1}(\omega)\right],$$

$$(1-\pi_t^j)\beta E_t^N[\Lambda_{t,t+1}Q_t K_t^{Bj}\bar{b}_t^j\Pi_{t+1}^{-1}] + \pi_t^j\beta\mathbb{E}_t^R[R_{t+1}^K Q_t K_t^{Bj}] \geq D_t^j. \tag{A.3}$$

[11] We also refer to Adrian and Shin (2014) and Nuño and Thomas (2017).

We define $\overline{D}_t^j = \overline{R}_t D_t^j$ and $\overline{b}_t^j = \left(\overline{R}_t D_t^j\right) / (Q_t K_t)$. As shown step-by-step in Rottner (2023), Appendix B, we can use a guess and verify approach to derive the incentive and participation constraint:

$$(1-\pi_t)\mathbb{E}_t^N\Big[\Lambda_{t,t+1}R_{t+1}^K(\theta\lambda_{t+1}+(1-\theta))\big(1-e^{\frac{-\psi}{2}}-\Omega_{t+1}\big)\Big] = \pi_t\mathbb{E}_t^R\Big[\Lambda_{t,t+1}R_{t+1}^K\big(e^{-\frac{\psi}{2}}-\overline{\omega}_{t+1}+\Omega_{t+1}\big)\Big]\,, \tag{A.4}$$

$$(1-\pi_t)\mathbb{E}_t^N\left[\beta\Lambda_{t,t+1}\overline{R}_t^D D_t\right] + \pi_t\mathbb{E}_t^R[\beta\Lambda_{t,t+1}R_{t+1}^K Q_t K_t^B\Big] = D_t\,. \tag{A.5}$$

Note that the incentive constraint and the participation constraint do not depend on intermediary j specific values. The multiplier for the incentive constraint (κ_t) and participation constraint (λ_t) are given as:

$$\kappa_t = \frac{\beta(1-\pi_t)\mathbb{E}_t^N\Lambda_{t,t+1}\left[\lambda_t-(\theta\lambda_{t+1}+1-\theta)\right]}{(1-\pi_t)\mathbb{E}_t^N\Lambda_{t,t+1}\left[(\theta\lambda_{t+1}+1-\theta)\,\widetilde{F}_{t+1}(\overline{\omega}_{t+1})\right]+\pi_t\mathbb{E}_t^R\Lambda_{t,t+1}\left[(\theta\lambda_{t+1}+1-\theta)\left(1-\widetilde{F}_{t+1}(\overline{\omega}_{t+1})\right)\right]}, \tag{A.6}$$

$$\lambda_t = \frac{(1-\pi_t)\mathbb{E}_t^N\Lambda_{t,t+1}R_{t+1}^K[\theta\lambda_{t+1}+(1-\theta)](1-\overline{\omega}_{t+1})}{1-(1-\pi_t)\mathbb{E}_t^N[\Lambda_{t,t+1}R_{t+1}^K\overline{\omega}_{t+1}]-\pi_t\mathbb{E}_t^R[\Lambda_{t,t+1}R_{t+1}^K]}. \tag{A.7}$$

A.2 Global Solution Method

We solve the model with global methods to account for the runs on the financial sector and the stochastic transition path to a net zero economy. We extend the global solution method of Rottner (2023), which can solve the type of run models studied here, to feature stochastic transition paths. Incorporating this feature is key to evaluating the financial stability impact of climate policies during the transition and in the long-run.

The state variables are previous period capital, current period net worth, the risk shock and the sunspot shock, that is $X_t = \left\{K_{t-1}, N_t, \epsilon_t^{\xi}, \epsilon_t^{\pi}\right\}$. We also condition the solution on the carbon tax path $\{\tau_l^c\}_{l=t}^{\infty}$ to allow for changing carbon taxes. In particular, we consider two distinct cases for the tax path in the model.

1. Long-run equilibrium: The tax path is constant. This case is relevant once the transition is completed or before agents anticipate future carbon taxation. The tax sequence is then constant for all periods, that is $\tau_t^c = \tau^c, \forall t$.

2. Transition dynamics: The tax path follows a commonly known path with time-varying taxes. Specifically, the tax path consists of two parts: First, the tax rate changes over time until its terminal level is reached in period T_{max}, that is $\{\tau_l^c\}_{l=t}^{T_{max}}$. This part reflects the transition. Once the maximum tax rate is reached, the tax remains constant, that is $\tau_t^c = \tau^c, \forall t > T_{max}$. At this level, we are back in the first case denoted as long-run equilibrium.

While the tax path is constant and known by the agents, the economy is subject to shocks. Therefore, we are analyzing stochastic transition dynamics, in which the agents expect the materialization of shocks.[12] The remaining parameters of the model are summarized as θ^P.

To find the model solution, we solve the policy functions for the asset price, consumption, the multiplier on the participation constraint, a measure of the promised repayments, and inflation. The policy function in period t depends on the state variables, the parameters, and the sequence of carbon shocks from period t onwards:

$$Q_t(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P), C_t(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P), \bar{b}_t(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P),$$
$$\Pi_t(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P), \quad \text{and} \quad \lambda_t(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P).$$

We also solve for the law of motion of net worth and the probability of observing a run next period:

$$N_t'(X_t, \epsilon_{t+1}^{\xi}; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P), \quad \text{and} \quad \pi_t(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P),$$

[12]The framework can be easily extended by tax paths that are subject to shocks themselves. In such a case, the shock component of the tax rate enters as a state variable.

where ϵ_{t+1}^{ξ} are the risk shock realizations next period. Once we have solved for these objects, we can back out all other variables.

The solution algorithm uses time iteration with linear interpolation (see, e.g., Richter, Throckmorton, and Walker, 2014). We also use an additional piecewise approximation of the policy functions by deriving separate policy functions to approximate the run and normal equilibrium. The expectations are approximated with Gauss-Hermite quadrature.

Our global solution algorithm is summarized below:

1. We first define a grid for the state variables (without the sunspot shock): $\boldsymbol{X} \in [\underline{K}_{t-1}, \overline{K}_{t-1}] \times [\underline{N}_t, \overline{N}_t] \times [\underline{\sigma}_t, \overline{\sigma}_t] \times [\underline{A}_t, \overline{A}_t]$. Using Gauss-Hermite quadrature, we set up integration nodes for the expectations with respect to the risk shock $\boldsymbol{\epsilon} \in [\underline{\epsilon^{\xi}}_{t+1}, \overline{\epsilon^{\xi}}_{t+1}]$. We denote the considered tax path by $\{\tau_l^c\}_{l=t}^{\infty}$.

2. We guess the piecewise linear policy functions to initialize the algorithm. This includes a separate guess for the policy function for each different carbon tax level. As an example, we have the following set of policy functions for the asset price:

$$\{Q_t(X_k; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P)\}_{t=t_0}^{\infty}$$

   While this would result in an infinite set of policy functions, we can exploit that policy functions are time-invariant once the terminal tax rate is reached. Therefore, we can simplify the set of policy functions that we need to solve for the two different cases that we consider:

   (a) Long-run equilibrium: The tax path is time-invariant. As a consequence, the policy function is not path-dependent. We then have:

$$Q_t(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P) = Q(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P) = Q(X_t; \tau^c, \Theta^P), \forall t$$

   (b) Transition dynamics: At the beginning, the tax path changes over time, while it then converges to the terminal rate. Therefore, we have two different problems at different stages in time:

$$\begin{aligned}
&\forall t \leq T_{max} : \{Q_t(X_t; \{\tau_l^c\}_{l=k}^{\infty}, \Theta^P)\}_{t=t_0}^{T_{max}} = \{Q_t(X_t; \{\tau_l^c\}_{l=k}^{T_{max}}, \Theta^P)\}_{t=t_0}^{T_{max}}, \\
&\forall t > T_{max} : Q_t(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P) = Q(X_t; \{\tau_l^c\}_{l=t}^{\infty}, \Theta^P) = Q(X_t; \tau_{T_{max}}^c, \Theta^P).
\end{aligned}$$

   This notation highlights that we can use a time iteration algorithm to solve for the policy functions. If we are in an infinite time horizon, that is either the long-run equilibrium or at the terminal tax rate, we solve the problem using policy function iteration. We can divide the transition dynamics in a finite horizon problem during the transition and an infinite horizon problem with the terminal tax rate after

$t > T_{max}$. We first solve the infinite horizon problem using policy function iteration as before. We can then use this to conduct backward induction to solve the finite horizon problem of the transition.

While we discussed this for the specific asset price policy function, this generalizes to all policy functions that we need to solve. In particular, we need a guess for all policy functions, the probability that a run occurs next period, and the law of motion of net worth. The latter $N_t'(X_t, \epsilon_{t+1}^{\xi}; \Theta^P)$ provides a mapping from state variables today into net worth next period at each integration point coming from Gauss-Hermite quadrature.

3. We solve for all time t variables for a given state vector and tax path ahead, focusing on the no run equilibrium. We use the law of motion for net worth and the run probability from the previous iteration j as given. We also need to calculate our next-period values using policy functions. In the infinite horizon problem, we use the guess of the policy function iteration from iteration $j-1$. In the finite horizon problem, we use the policy function from period $t+1$, i.e. $Q_{t+1}(\cdot)$. Expected values are computed using Gauss-Hermite quadrature. We then use a numerical root finder to minimize the error of these equations. The inputs are the time-invariant policy functions in the infinite horizon problem and the period t policy functions for the finite horizon problem:

$$\begin{aligned}
\text{err}_1 &= \left(\frac{\Pi_t}{\Pi} - 1\right)\frac{\Pi_t}{\Pi} - \left(\frac{\epsilon}{\rho^r}\left(MC_t - \frac{\epsilon-1}{\epsilon}\right) + \Lambda_{t,t+1}\left(\frac{\Pi_{t+1}}{\Pi} - 1\right)\frac{\Pi_{t+1}}{\Pi}\frac{Y_{t+1}}{Y_t}\right), \\
\text{err}_2 &= 1 - \beta \mathbb{E}_t \Lambda_{t,t+1}\frac{R_t^I}{\Pi_{t+1}}, \\
\text{err}_3 &= (1-\pi_t)\mathbb{E}_t^N\left[\beta\Lambda_{t,t+1}\overline{R}_t D_t\right] + \pi_t \mathbb{E}_t^R\left[\beta \Lambda_{t,t+1} R_{t+1}^K Q_t K_t^B\right] - D_t, \\
\text{err}_4 &= (1-\pi_t)\mathbb{E}_t^N\left[\Lambda_{t,t+1}R_{t+1}^K(\theta\lambda_{t+1} + (1-\theta))(1 - e^{\frac{-\psi}{2}} - \Omega_{t+1})\right] \\
&\qquad - \pi_t\mathbb{E}_t^R\left[\Lambda_{t,t+1}R_{t+1}^K(e^{-\frac{\psi}{2}} - \overline{\omega}_{t+1} + \Omega_{t+1})\right], \\
\text{err}_5 &= \lambda_t - \frac{(1-\pi_t)\mathbb{E}_t^N\Lambda_{t,t+1}R_{t+1}^K[\theta\lambda_{t+1} + (1-\theta)](1-\overline{\omega}_{t+1})}{1 - (1-\pi_t)\mathbb{E}_t^N[\Lambda_{t,t+1}R_{t+1}^K\overline{\omega}_{t+1}] - \pi_t\mathbb{E}_t^R[\Lambda_{t,t+1}R_{t+1}^K]}.
\end{aligned}$$

4. We now take our policy functions as well as the law of motion for net worth and the run probability from iteration $j-1$ as given. Using these objects, we calculate the variables for the period t and $(t+1)$ variables. We use these points to calculate N_{t+1} across the integration nodes and update the law of motion for net worth:

$$N_{t+1} = \max\left[R_{t+1}^K Q_t K_t^B - \overline{R}_t D_t, 0\right] + (1-\theta)\zeta K_t. \tag{A.8}$$

To determine whether the run equilibrium is supported on a specific node, we compute

$$R^K_{t+1}Q_tK^B_t - \overline{R}_tD_t \leq 0. \tag{A.9}$$

which evaluates to one if a run is possible.[13] This can be now used to evaluate the probability of a run next period based on Gauss-Hermite quadrature.

5. We repeat the steps 3 and 4 focusing also on the run equilibrium in the current period.

6. We update the policy functions, the law of motion for net worth, and the run probability using a weighted combination of our results from iteration j and the previous guess.

7. We repeat the steps 3 - 6 until the errors of all functions (policy functions, law of motion of net worth, and the probability of a run) are sufficiently small at each point of the discretized state space.

8. The infinite horizon problem is solved at this stage.

   For the finite horizon problem, we redo the steps 3 - 7 for all periods backwards, i.e. in the order of $T_{max}-1, T_{max}-2, ..., t_0+1$, and finally t_0. Furthermore, we also need to calculate one additional object. The law of motion that gives the mapping from the period t_0-1 without the tax path and the period in which the transition arrives unanticipated: $N'_{t_0-1}(X_{t_0-1}, \epsilon^{\xi}_{t_0}; \{\tau^c_l\}^{\infty}_{l=t_0-1}, \Theta^P)$. For this, we use step 4 using the old policy functions for the period $t_0 - 1$ (the previous long-run equilibrium) and the newly obtained policy functions for period $t_0 - 1$.

[13] The equation implies a zero and one indicator, which is a very unsmooth object. As a consequence, we use the following functional forms based on the exponential function to determine the run probability in this state of the world: $\frac{\exp(\zeta_1(1-D_{t+1}))}{1+\exp(\zeta_1(1-D_{t+1}))}$ where $D_{t+1} = \frac{R^k_{t+1}}{R^D_t}\frac{\phi}{\phi-1}$ at each calculated N_{t+1}. We set ζ_1 to 500 which ensures sufficient steepness. While this approximation induces smoothness, it is still very close to an indicator function with 0 and 1 values.

Table A.1: Targeted Moments

Moment	Data	Model
Mean leverage (p.p.)	15.5	15.5
Standard deviation leverage (p.p.)	3.0	3.0
Autocorrelation leverage	0.96	0.95
External finance premium	3.0	3.0
Share of assets held by intermediaries	33%	35%
Crisis probability	2% p.a.	2.1% p.a.

A.3 Calibration

Since the carbon tax is expressed in abstract model units, which are hard to interpret, we transform the tax rate into carbon prices, i.e., dollars per unit of emissions. We follow the literature (Ferrari and Nispi Landi, 2023) and relate model-implied output y_t and emissions e_t in the initial stationary distribution to current world GDP ($y^{world} =$ 105 trillion USD at PPP in 2023) and current global carbon emissions ($e^{world} = 37.5$ GtCO2 in 2023), respectively. Since output and emissions are normalized to one in the initial stationary distribution, the carbon price in dollars per tonne of carbon ($/tCO2) associated with a given tax τ_t^c is then given by $p_t^c = \frac{y^{\text{world}}}{e^{\text{world}}}\tau_t^c$. Under the 2050 value of the abatement cost function $b_{1,2050}$, we obtain a net zero tax of $\tau_t^c = 0.03$ from (1). This corresponds to a carbon price of 95$/tCO2.[14]

We can now use the obtained functions to simulate the model's ergodic distribution and transition dynamics. In Table A.1, we demonstrate that our baseline calibration without carbon taxes is able to fit key moments in the data well.

[14]While this tax appears quite small compared to currently observed emission permit prices in the EU emission trading scheme, it has to be noted that *all* emissions are taxed in our macroeconomic model. In contrast, only a limited share of emissions is subject to emission trading schemes or carbon taxes and firms receive a considerable amount of free allowances in practice.

B Additional Quantitative Results

B.1 Non-Linear Effect of Carbon Taxes

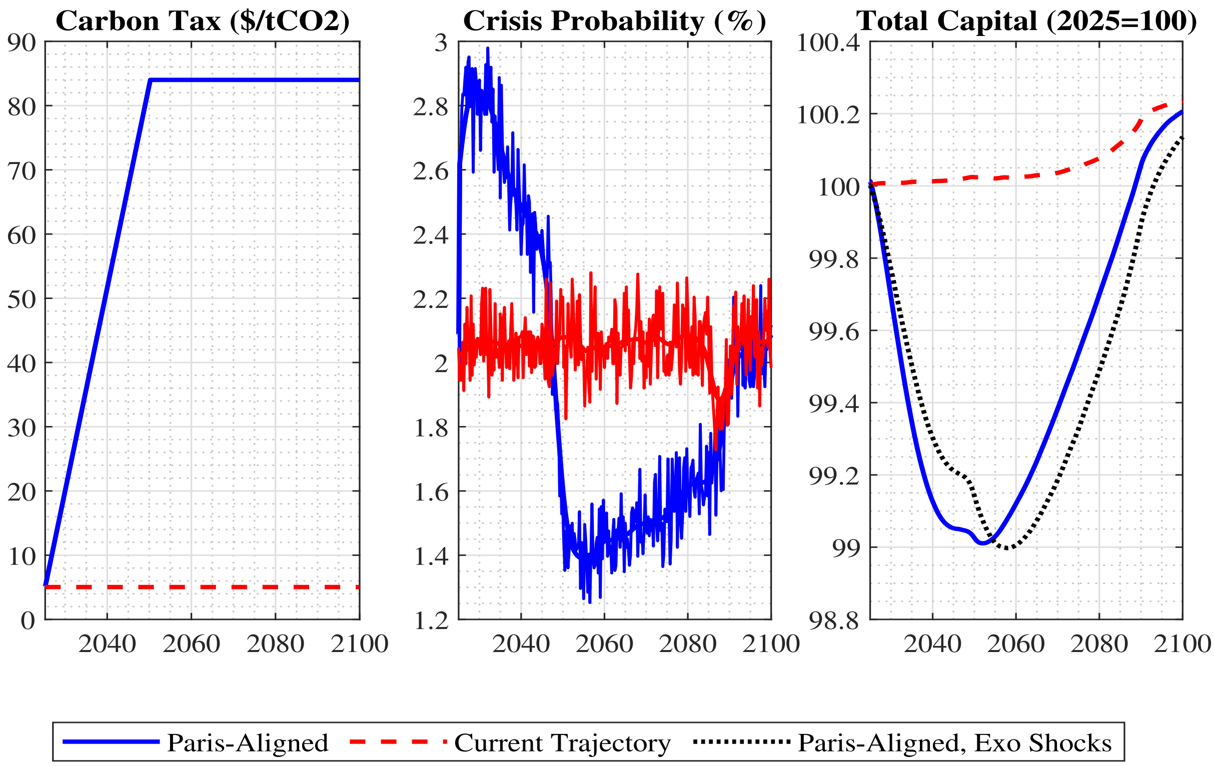

Figure B.1: Transition path to net zero, additional variables. The dynamics are shown for the years 2024Q4 until 2100Q4. The results are based on simulating the model 100,000 times with a burn-in period of 200 quarters.

In the middle panel of Figure B.1, we display the run probabilities without the cubic spline smoothing used in Figure 1. The right panel illustrates that the decline in total capital predominantly stems from the climate policy-induced wedge in the marginal product. While the solid blue line represents the actual transition to higher carbon taxes, the dotted black line corresponds to the simulation using the run realizations from the *current trajectory*, as described in Section 4.3. Since there are *fewer* runs during the first periods of the transition, the capital stock is slightly larger on the counterfactual path than on the actual transition path. This pattern reverts during later stages of the transition, which are characterized by comparatively *more* runs on the *current trajectory*. Quantitatively, the difference is relatively small compared to the role of the carbon tax-induced wedge in the return on capital.

To illustrate the workings of our non-linear model, Figure B.2 displays the impulse response functions of key variables to a positive two-standard-deviation shock to exogenous risk ξ_t at different stages of the net zero transition. The solid black line refers to the case without carbon tax, the dashed red line is eight quarters into the transition, and the dotted green line refers to the terminal equilibrium with net zero emissions.

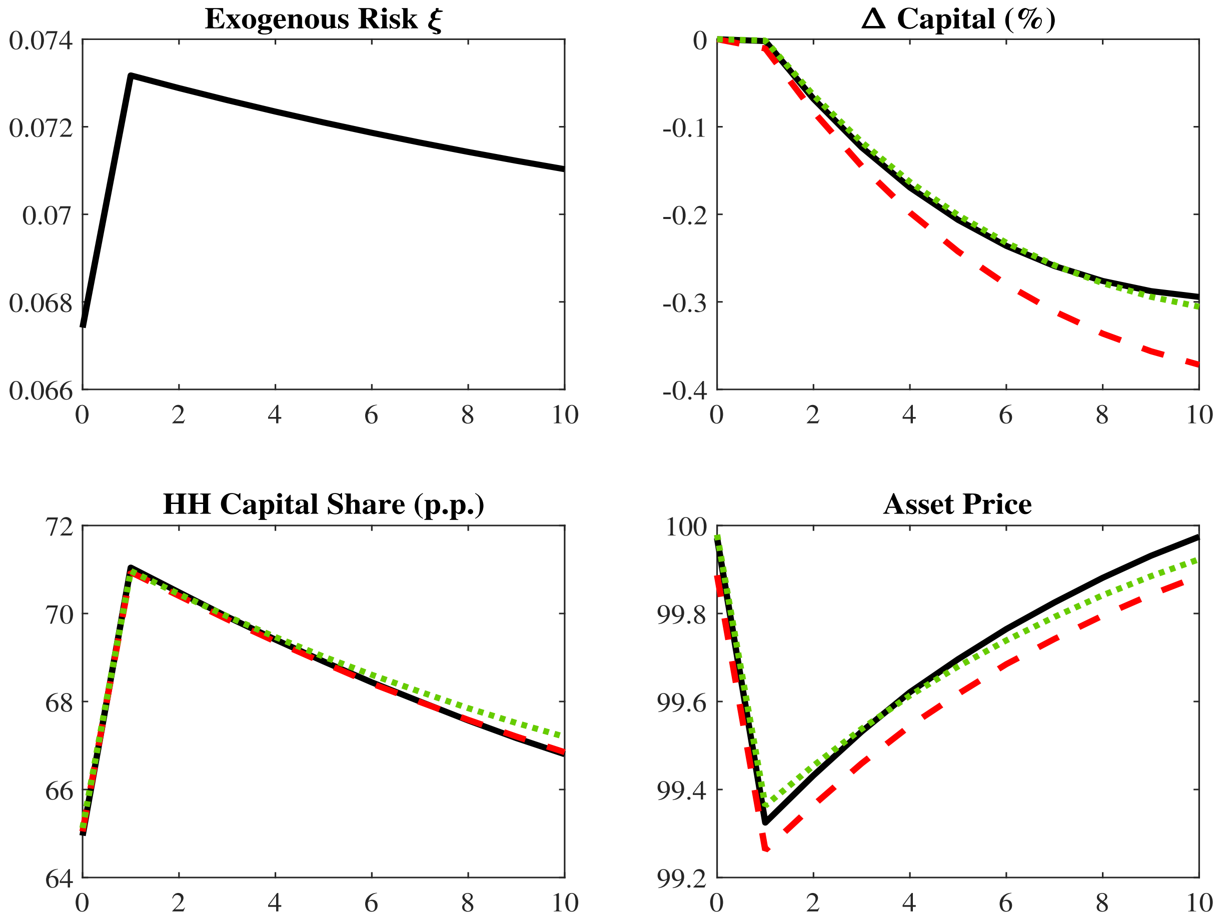

Figure B.2: Impulse response functions. This figure displays impulse responses to a positive two-standard-deviation risk shock at different stages of the transition. The solid black line refers to the case without carbon tax, the dashed red line is eight quarters into the transition, and the dotted green line refers to the terminal equilibrium with net zero emissions. The IRFs are obtained by simulating the model 10.000 times after a burn-in of 1000 quarters. The change in capital is expressed relative to the period before the shock. The household capital share is in percentage points. The asset price is normalized to 100 in the long-run mean.

By increasing risk-shifting incentives, this shock tightens intermediaries' leverage constraint, which is associated with a contraction in credit supply, economic activity, and investment. The upper right panel shows this specifically for total capital, which persistently declines by up to 0.4 percent. Deleveraging is achieved by selling some capital to households, who hold more than 70% of the capital stock, compared to 65% in the long-run mean. Importantly, the increase in household capital holdings is almost identical, irrespective of the stage of the transition. By contrast, the asset price response is quite different. The drop is much more pronounced shortly after the shift onto the *Paris-aligned* path, as capital holdings are too large in light of the now smaller productivity of capital. As a consequence, households are only willing to hold capital at a lower price, reflected by the dotted red line in the bottom left panel of Figure B.2. This reinforces the pressure on intermediaries to sell, which further depresses investment and also increases the run probability.

Notably, the long-run equilibrium with net zero emissions experiences a slightly smaller asset price drop to the same shock, which is in line with the smaller crisis probability in the long run. These dynamics illustrate how a sudden shift from one monotonically increasing tax path to a steeper, but still monotonic, tax path can have non-monotonic effects on the crisis probability.

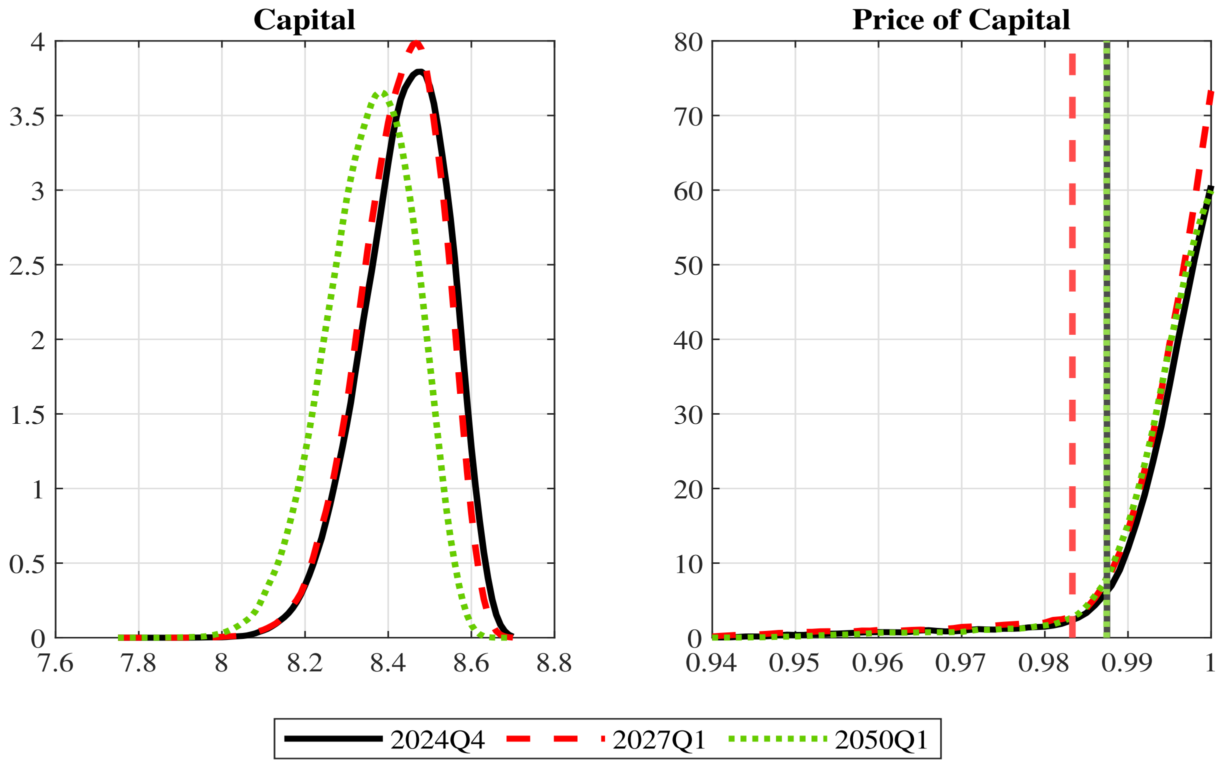

Figure B.3: Non-Linearities during the transition to net zero. This figure shows the distribution of endogenous model objects at different stages of the Paris-aligned transition. The distributions are obtained by simulating the model 100,000 times with a burn-in period of 200 quarters. Capital holdings are expressed relative to quarterly GDP. The 5%-quantiles are indicated by vertical lines for the asset price.

In Figure B.3, we show the distribution of total capital and the price of capital at different points of the transition path. The last quarter on the *current trajectory* is indicated by solid black lines. Dotted red lines represent the distribution 10 quarters into the transition, while the dotted green lines correspond to the first quarter at which taxes reach net zero. The left panel shows total capital, which declines from its initial distribution in a quite monotonic fashion towards the long-run equilibrium. In sharp contrast, the non-linear implications of the transition are represented by the price of capital in the right panel. Focusing on the 5% quantiles, indicated by vertical lines, we observe a more pronounced tail ten quarters into the transition (dashed red line), while the tail features less mass in the new long-run equilibrium (dotted green line).

B.2 Climate Policy, Capital Accumulation and Fragility

Figure B.4 demonstrates how varying carbon taxes affect the macroeconomy and financial stability in the long run. We fix the abatement cost slope at its 2025 value $b_{1,2025} = 0.05$ and solve the model for different time-invariant carbon tax levels. We consider values ranging from zero to 140\$/tCO2, which implies full abatement for the 2025 value of the abatement cost parameter $b_{1,2025} = 0.05$.

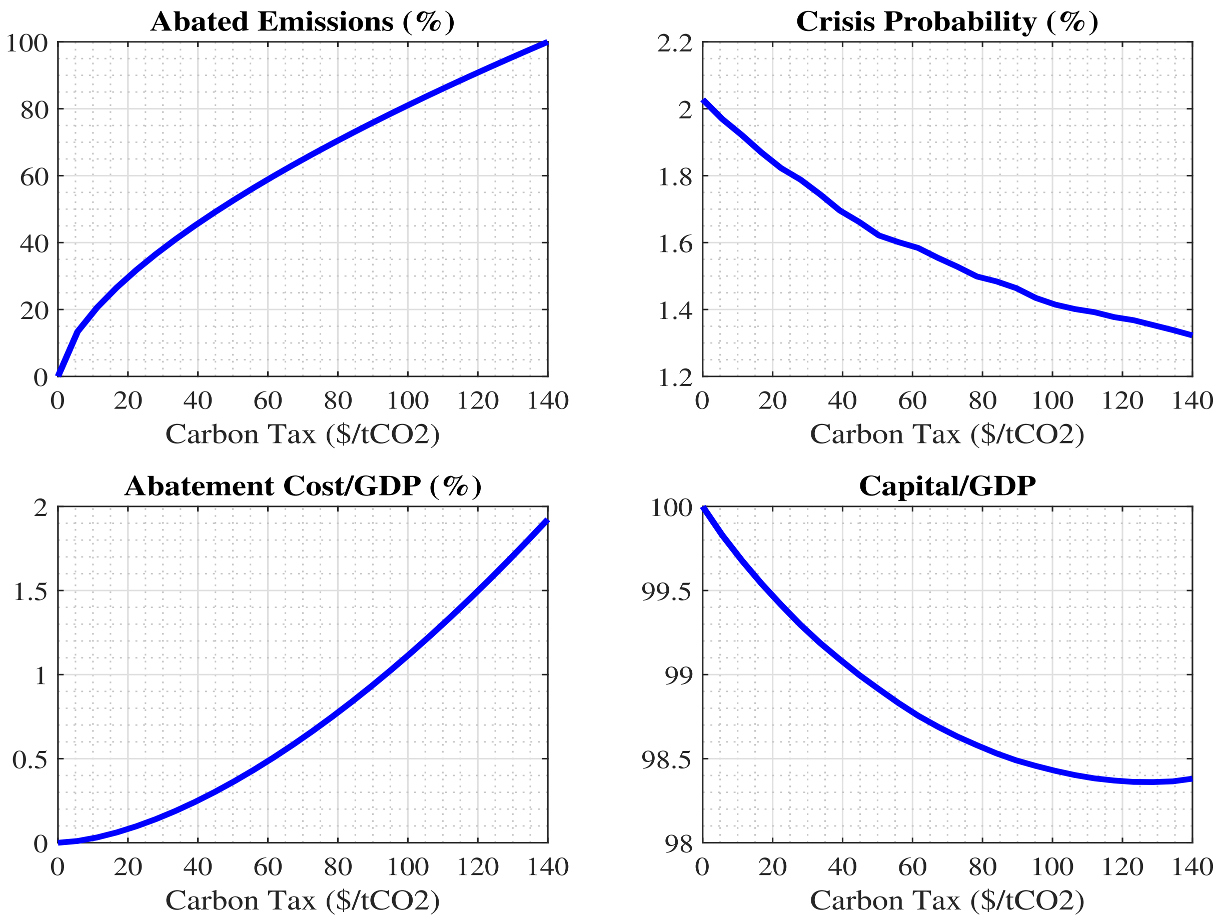

Figure B.4: Carbon taxes and financial stability in the long-run. The impact of varying carbon taxes on abated emissions, the annualized crisis probability, the share of abatement costs to GDP, and the capital/GDP ratio. The capital/GDP ratio is indexed to 100 for the case without carbon taxes. The green dotted line indicates the status quo carbon tax of 2023. The results are based on simulating the model for 100,000 periods with a burn-in of 10,000 periods.

The upper left panel of Figure B.4 shows the share of abated emissions implied by a given long-run carbon tax. The bottom left panel shows that abatement costs increase in a concave fashion towards full abatement. In the bottom right panel, we demonstrate that higher carbon taxes are associated with less capital accumulation as measured by the capital/GDP ratio. For taxes exceeding 100 \$/tCO2, this ratio is around 1.5% smaller than in the long-run equilibrium without carbon taxes.

The top right panel of Figure B.4 reveals that the annualized crisis probability declines from around 2% to around 1.4% under the 140%/tCO2 tax consistent with full abatement.

This follows from an equilibrium effect operating through capital accumulation and the relative size of the financial sector. Since carbon taxes reduce the average productivity of the economy, aggregate capital is smaller in the long run. This depresses the crisis probability, since households have to acquire less capital if financial intermediaries need to sell assets to reduce their leverage ratio. It follows from their period utility function (8) that they are willing to pay a higher price for holding capital, ceteris paribus. Thus, intermediaries are more likely to be able to service depositors, which reduces the partition of the state space supporting the run equilibrium, thereby reducing the run frequency.

B.3 Model Modifications

In this section, we provide more details on some model extensions and climate policy variations presented in Section 5.

Abatement Subsidies In this extension, we change the underlying assumption regarding the use of carbon tax revenues. So far, we assumed that carbon tax revenues are rebated to households in a lump sum fashion. Now, the carbon tax revenue is used to subsidize firms' adoption of clean technologies. The subsidy is financed with the carbon tax revenue so that the per-unit subsidy is given by $((1-\eta_t^*)\tau_t^c)/\eta_t^*$. The share of clean technology firms η_t^* enters the technology choice problem of each individual firm, as they take the size of the subsidy as exogenously given. The subsidy effectively reduces the cost of operating the clean technology to $b_{1,t}\eta_t^{b_2} - ((1-\eta_t^*)\tau_t^c)/\eta_t^*$ and the indifference point in equilibrium is given by

$$\eta_t^* = \min\left\{ \left(\frac{\tau_t^c}{b_{1,t}}\right)^{\frac{1}{b_2+1}}, 1 \right\}, \tag{B.1}$$

where we have used the additional equilibrium condition $\eta_t = \eta_t^*$. Note that the share of clean firms with abatement subsidies is larger than in the baseline model, which is given by equation (1), for any given carbon tax. The resource costs do not change relative to the baseline specification. However, the wedge in the first-order condition of investment is smaller for each tax level than under rebates to households. Consistent with the baseline model, we compute the transition dynamics with abatement subsidies for the *Paris-aligned* tax path consistent with net zero in 2050.

Physical Climate Damages Figure B.5 displays the effects of shifting to an ambitious carbon tax path if climate damages reduce TFP rather than inflicting utility losses on households. The bottom left panel illustrates that the initial financial stability impact of shifting to the *Paris-aligned* tax path is remarkably similar to the baseline model. Likewise, the crisis probability drops below the current trajectory during later stages of the transition. Different from the baseline model, however, the long-run crisis probability remains slightly above the *current trajectory*. Since physical climate damages further depress productivity and capital accumulation, they in turn diminish the demand for financial services and the size of the financial sector. Taken together, the welfare impact associated with "Climate Minsky Moments" is negative.[15]

[15] It should be noted that the exact welfare decomposition into productivity losses from emission abatement and temperature gains is not feasible in this extension. This follows because climate damages are no longer entering the welfare function in an additive fashion through households' utility, but directly interact with output and, thereby, consumption, labor, and capital accumulation. Hence, we only report the net welfare effect of ambitious climate policy in Table 2.

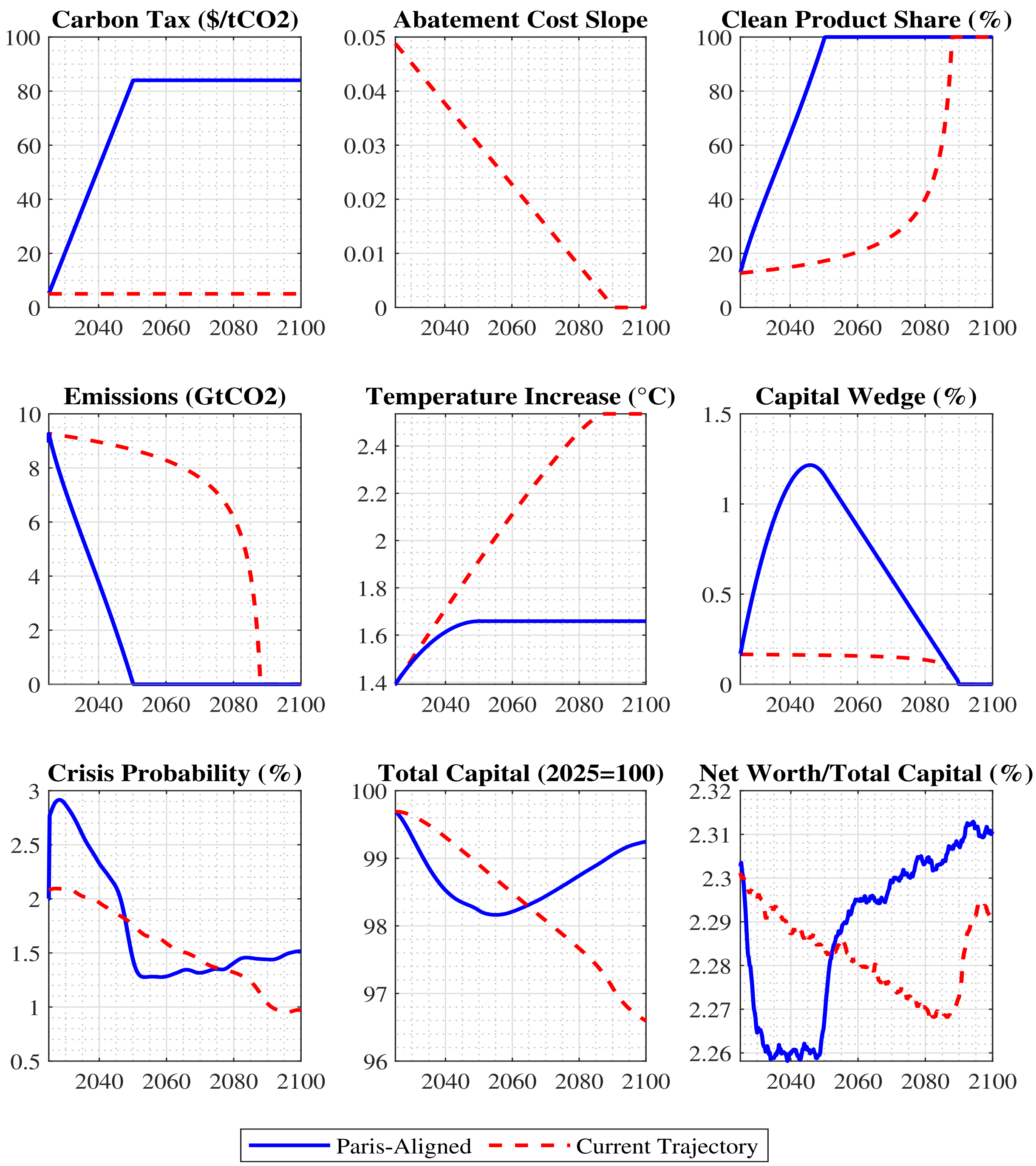

Figure B.5: Transition path to net zero with physical climate risk. Comparison of the impact of alternative transition paths - Paris-aligned, current trajectory, and policy freeze - on selected variables. The dynamics are shown for the years 2024Q4 until 2100Q4. The results are based on simulating the model 100,000 times with a burn-in period of 200 quarters. Crisis probabilities are annualized, and cubic spline smoothing is applied to remove sampling error for improved readability.

B.4 Welfare

This section presents additional results on the welfare implications of "Climate Minsky Moments".

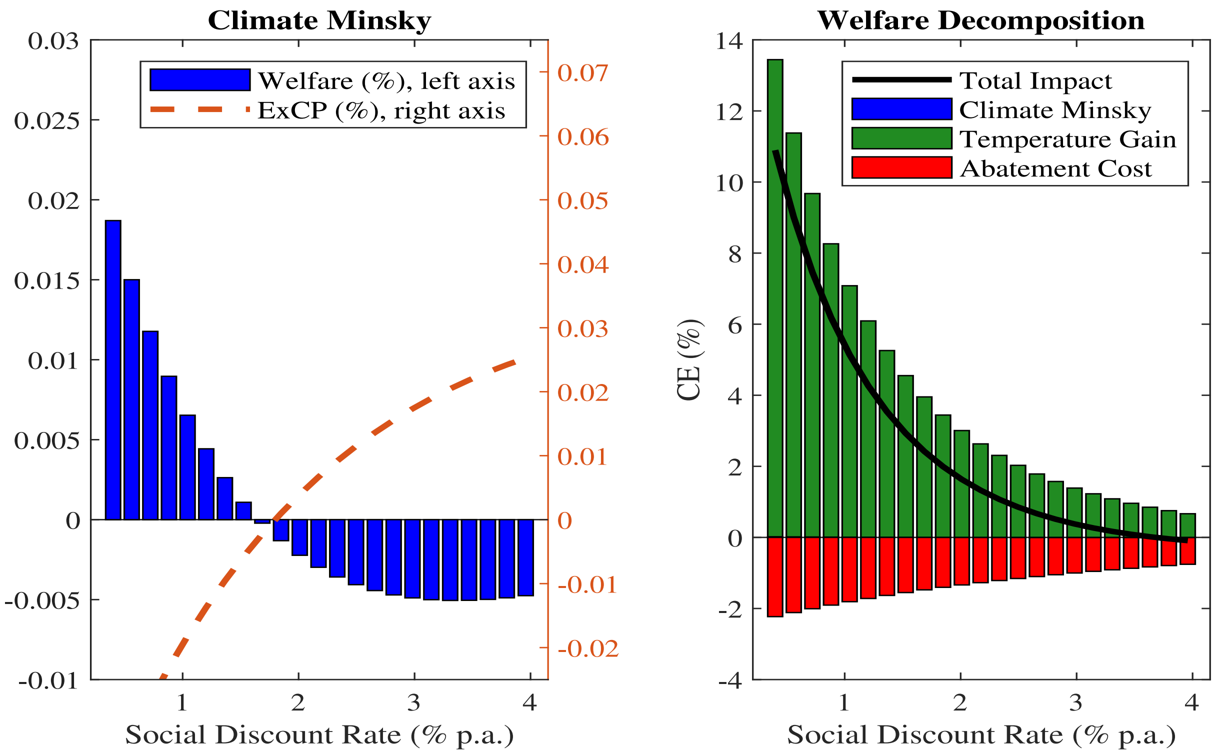

Figure B.6: Welfare and financial stability effects without households' capital management costs. Left panel: Welfare effect of "Climate Minsky Moments" (left axis) and the excess crisis probability (right axis) for varying social discount rates. Right panel: Decomposition of total welfare into contributions from "Climate Minsky Moments", temperature gains, and productivity losses. The results are obtained from simulating the model 100,000 times with a burn-in period of 200 quarters.

Excluding Capital Management Cost In order to alleviate the concern that the welfare implications of "Climate Minsky Moments" are primarily driven by households' cost of managing physical capital, we decompose welfare into temperature gains, productivity losses, and financial stability effects, but exclude capital management costs from the welfare function. Importantly, these costs still affect the economy along the transition through households' first-order conditions. The results are displayed in Figure B.6. The picture is overall very similar to the baseline welfare decomposition (Figure 2), although the magnitude of the financial stability effect in the left panel is a bit lower. For instance, under very low social discount rates, the financial stability gains correspond to 0.025% in CE when capital management costs are excluded, compared to 0.03% in the baseline. By contrast, the welfare losses shrink to 0.005% for a social discount rate of 4%, compared to -0.008% in the baseline. Importantly, the point at which the welfare effect switches signs is hardly affected by the exclusion of capital management costs.

The Role of the Financial Cycle The state of the economy at the onset of the transition plays an important role for the impact of climate policies on financial stability and welfare. While figure 3 displays the effect on the crisis probability, we show in this section how the financial sectors' loss-absorbing capacity affects the excess crisis probability and the welfare losses from "Climate Minsky Moments".

To obtain an appropriate comparison, we impose the same risk shock realizations on the *current trajectory* and display the results in Figure B.7. Comparing the left panel (low risk) to the right panel (high risk), it stands out that the excess crisis probability is consistently larger if the economy's loss-absorbing capacity is small. Consequently, the welfare effect is also smaller for every social discount rate and turns negative at about 1%, while it is only negative for rates exceeding 2% if the loss-absorbing capacity is large. This observation is an outcome of our non-linear model, which shows that the financial stability effects of ambitious climate policy are state-dependent in a non-linear fashion.

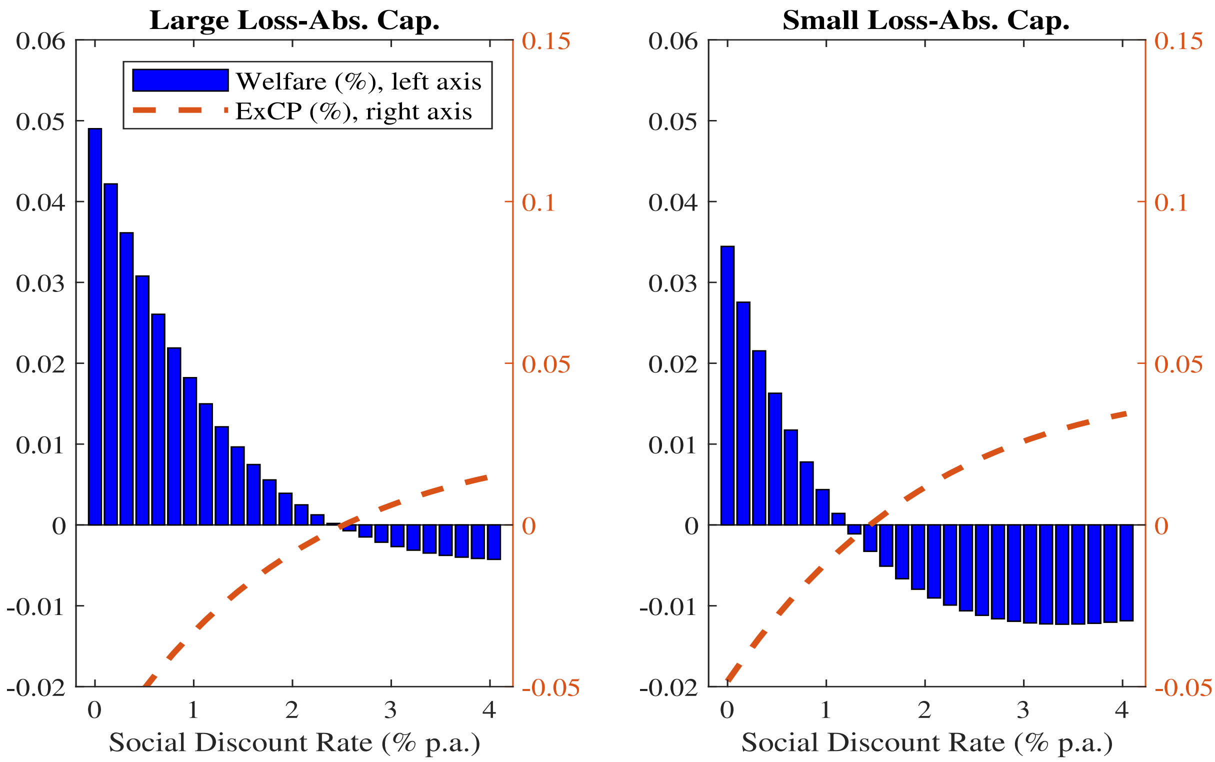

Figure B.7: The Role of the Financial Cycle. Welfare Effects of "Climate Minsky Moments" (left axis) and the excess crisis probability (right axis) for varying social discount rates. The left panel corresponds to the low-risk case with a large loss-absorbing capacity by intermediaries. The right panel corresponds to the high-risk case with a small loss-absorbing capacity. The results are obtained from simulating the model 100,000 times with a burn-in period of 200 quarters.

The Stance of Monetary Policy Similar to the financial sector's loss-absorbing capacity, the stance of monetary policy also affects the likelihood of "Climate Minsky Moments". While we have compared crisis probabilities for different monetary policy stances under the same Paris-aligned carbon tax path in figure 4, we show how the monetary policy stance affects the excess crisis probability and the welfare effect of "Climate Minsky Moments" in Figure B.7. The welfare losses are generally more pronounced if monetary policy is contractionary in the early stages of the transition, as this puts further pressure on intermediaries. By contrast, the net welfare effect of "Climate Minsky Moments" is positive even for fairly high social discount rates if monetary policy is expansionary.

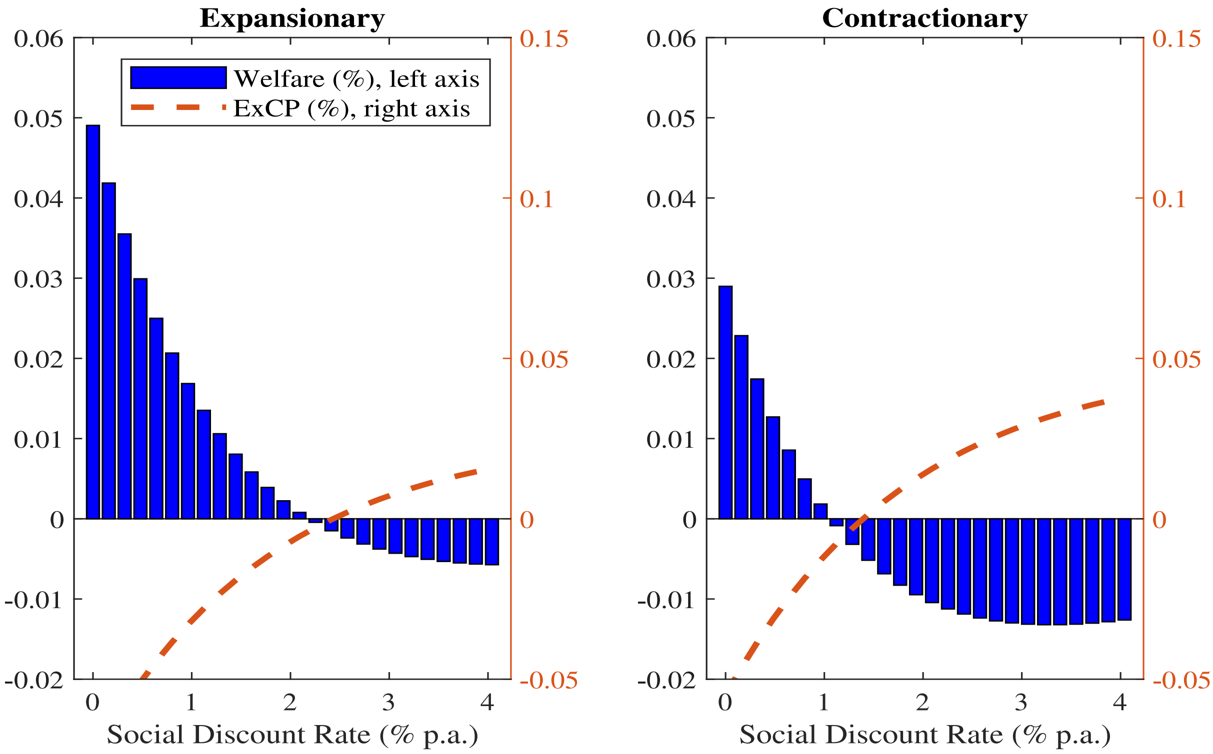

Figure B.8: The Role of Monetary Policy. Welfare Effects of "Climate Minsky Moments" (left axis) and the excess crisis probability (right axis) for varying social discount rates. The results are obtained from simulating the model 100,000 times with a burn-in period of 200 quarters.